Purdue University College of Engineering
Formation Study

Review 5.31.18

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Agenda

1. Program Update
2. Site Planning Opportunities
3. Option 1: The Quad, phases 1,2,3 & 4
4. Option 2: The Serpentine, phases 1,2,3 & 4
5. Architectural Context and Opportunities

1. Program Update

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2. Site Planning Opportunities

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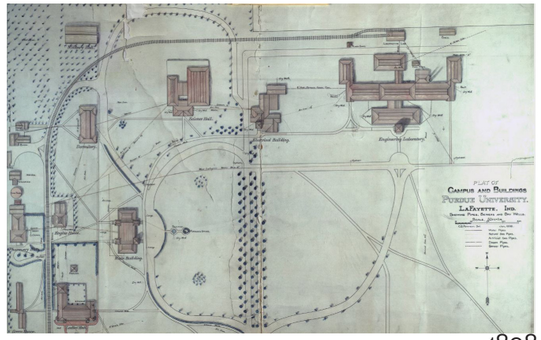
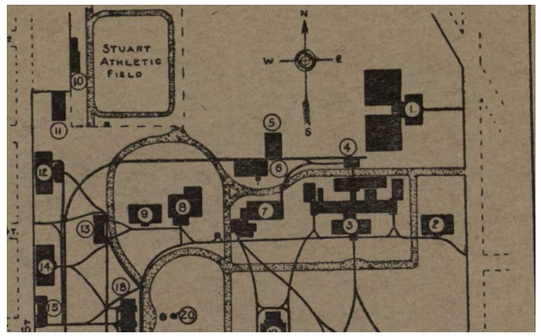
1. Site Planning Opportunities: Existing Framework



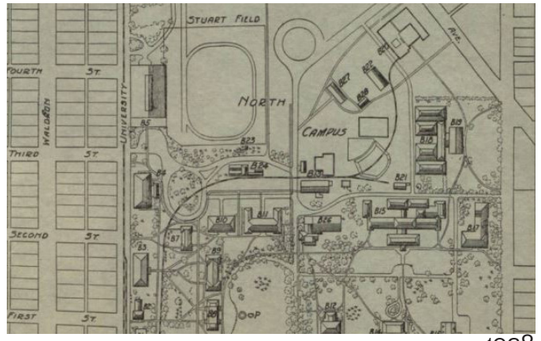
1898

1955



1910

1968



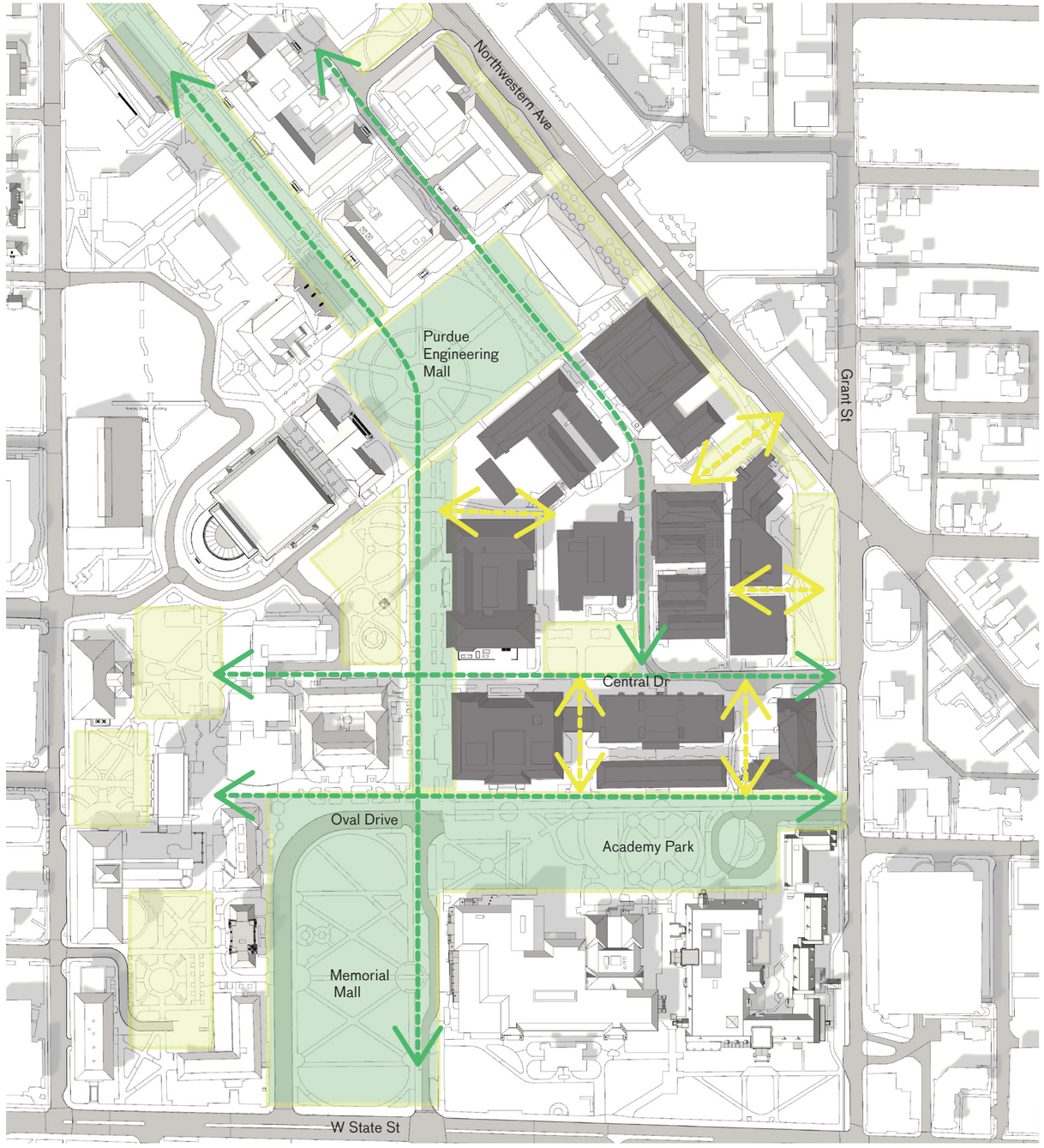
1928

Mall

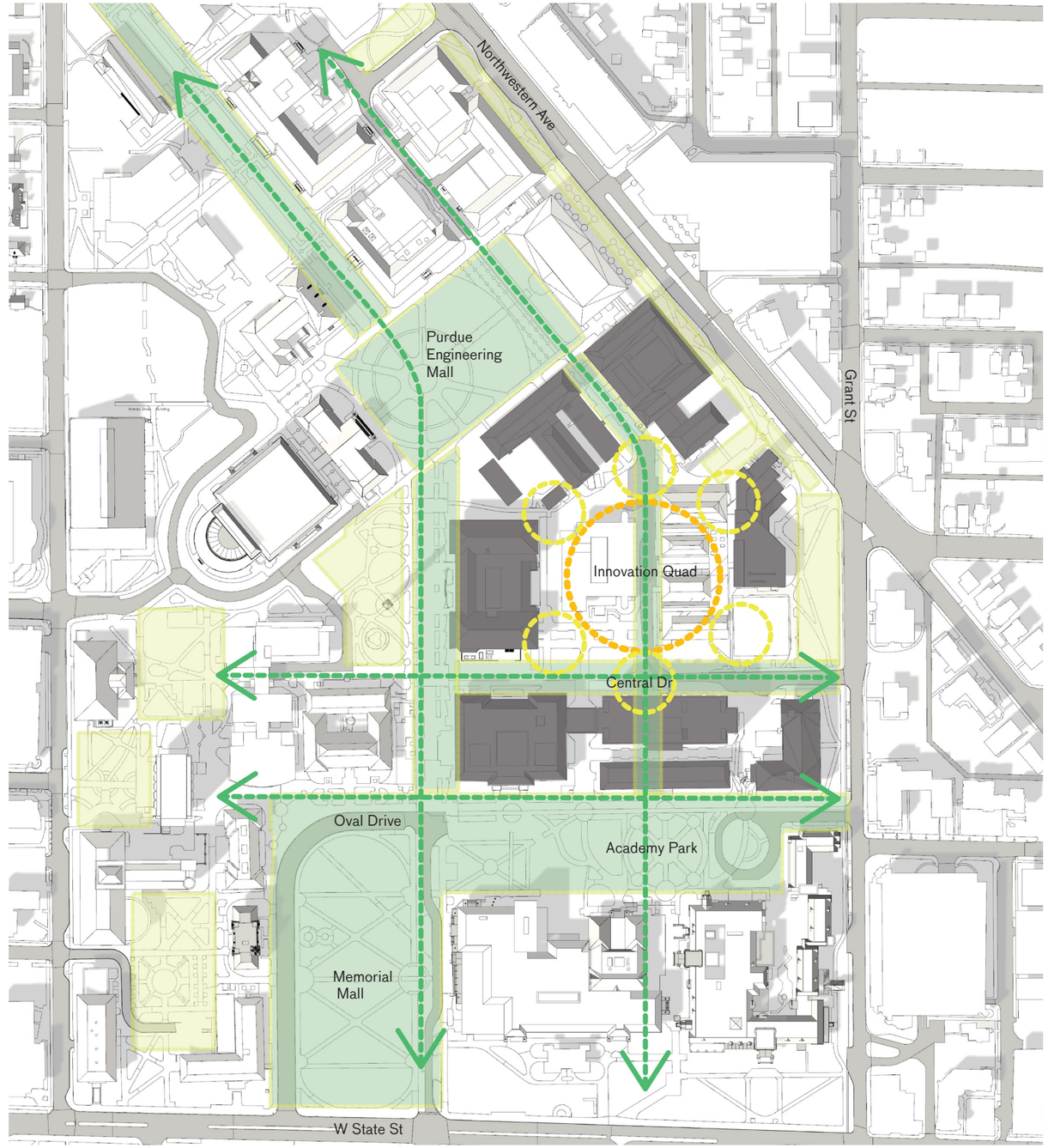
Secondary public space

Primary pedestrian route

Secondary pedestrian connection



1. Site Planning Opportunities: A New Innovation Quad



Mall

Secondary public space

Primary pedestrian route

Threshold

New campus open space

3. Massing Option 1: The Quad

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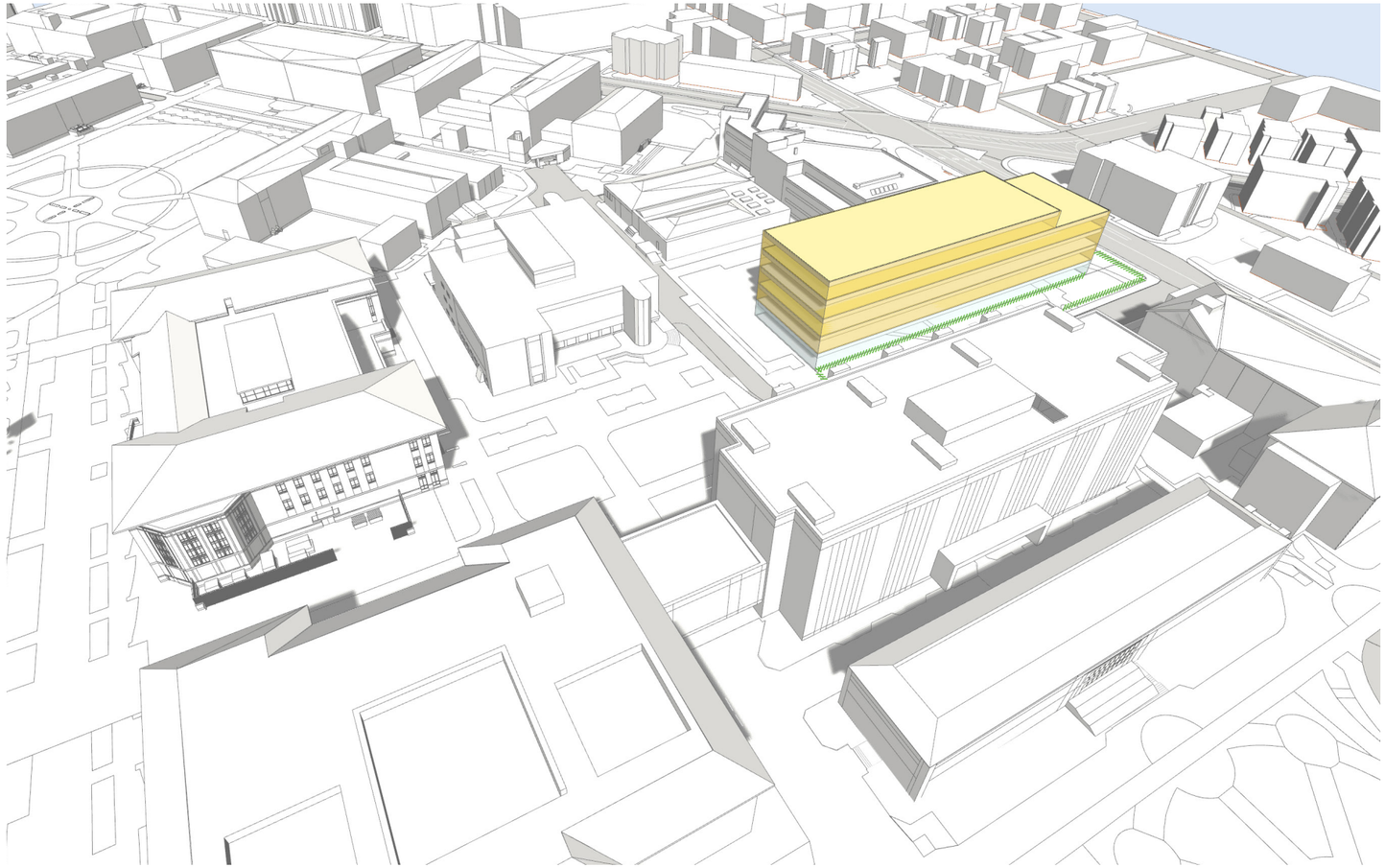
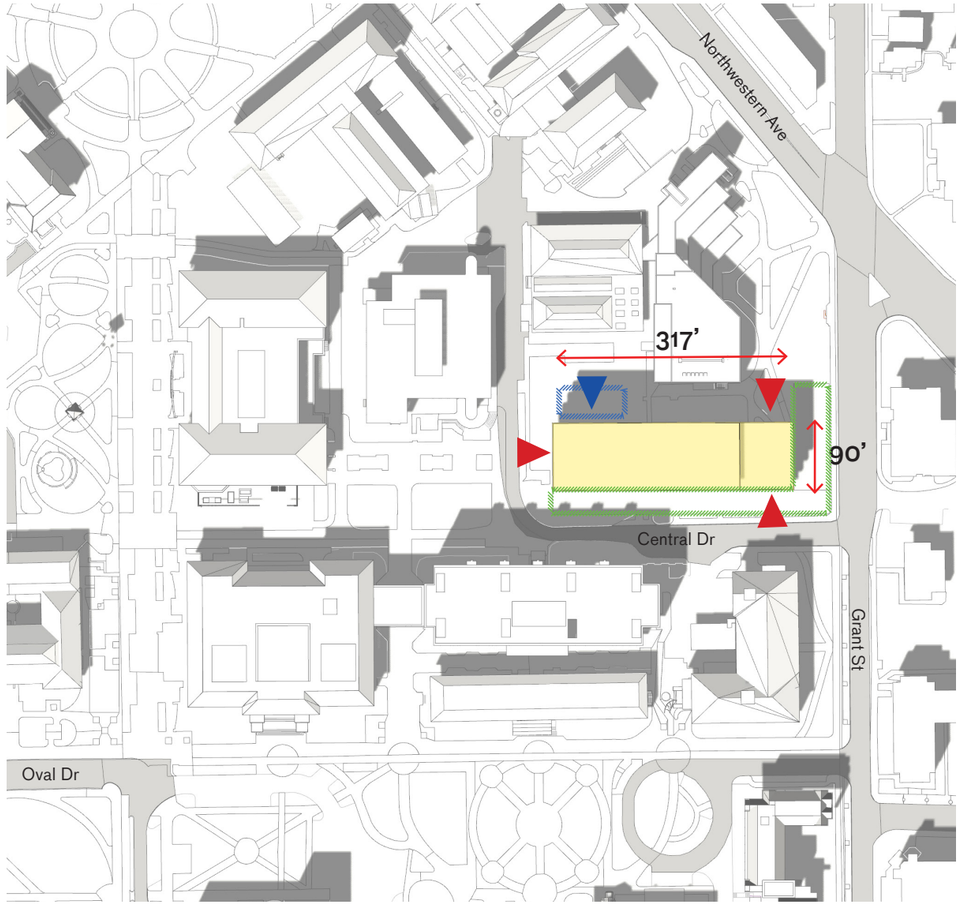
Quad: Phase 1





Floors: B+5
GSF: 165,083

B	28,575
1	28,575
2	28,575
3	28,575
4	28,575
5	22,208
6	
Totals	**165,083**

1 Part demolition of MGL

2 Temporary Loading Established off Central Drive

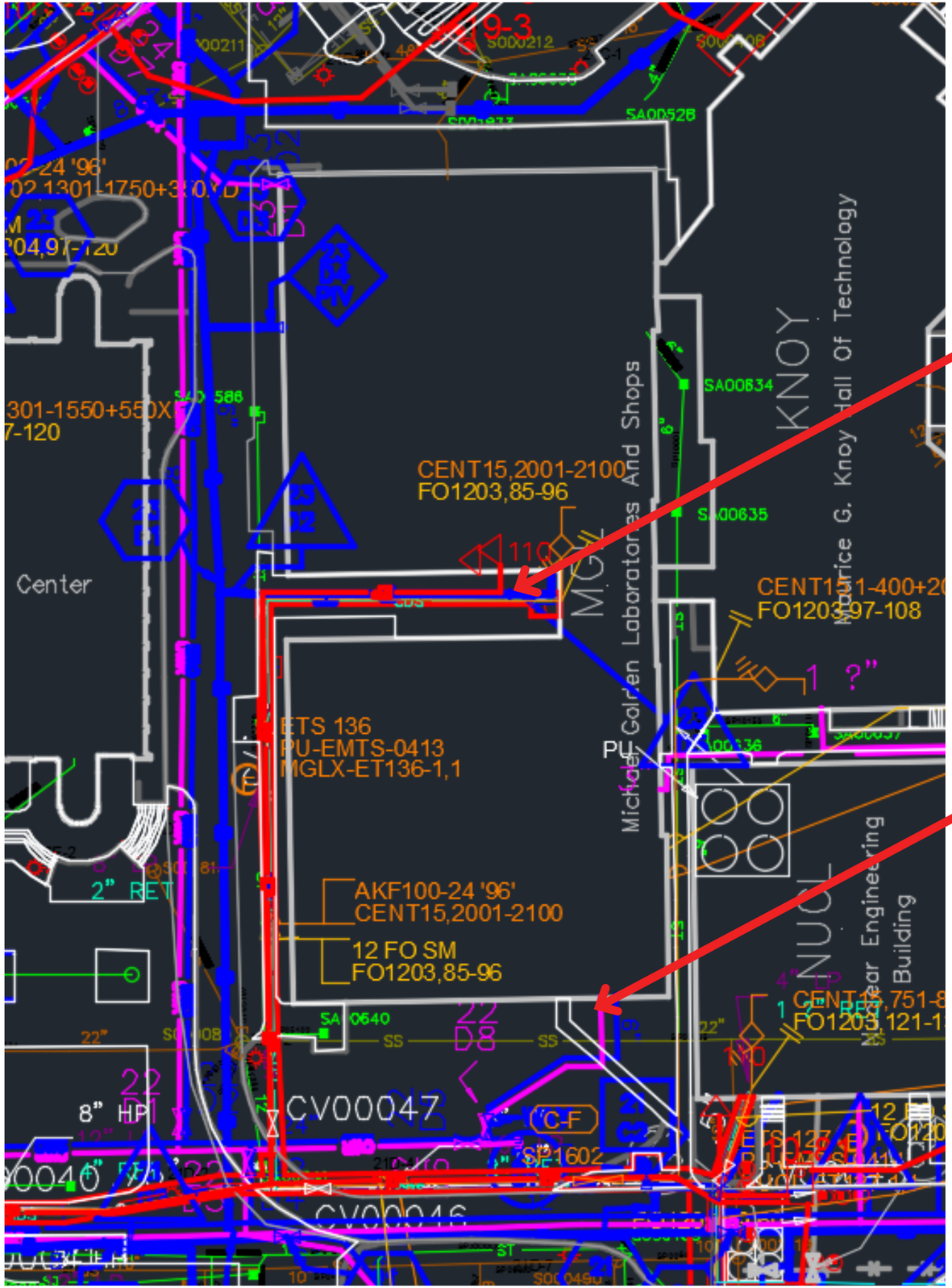
Loading dock

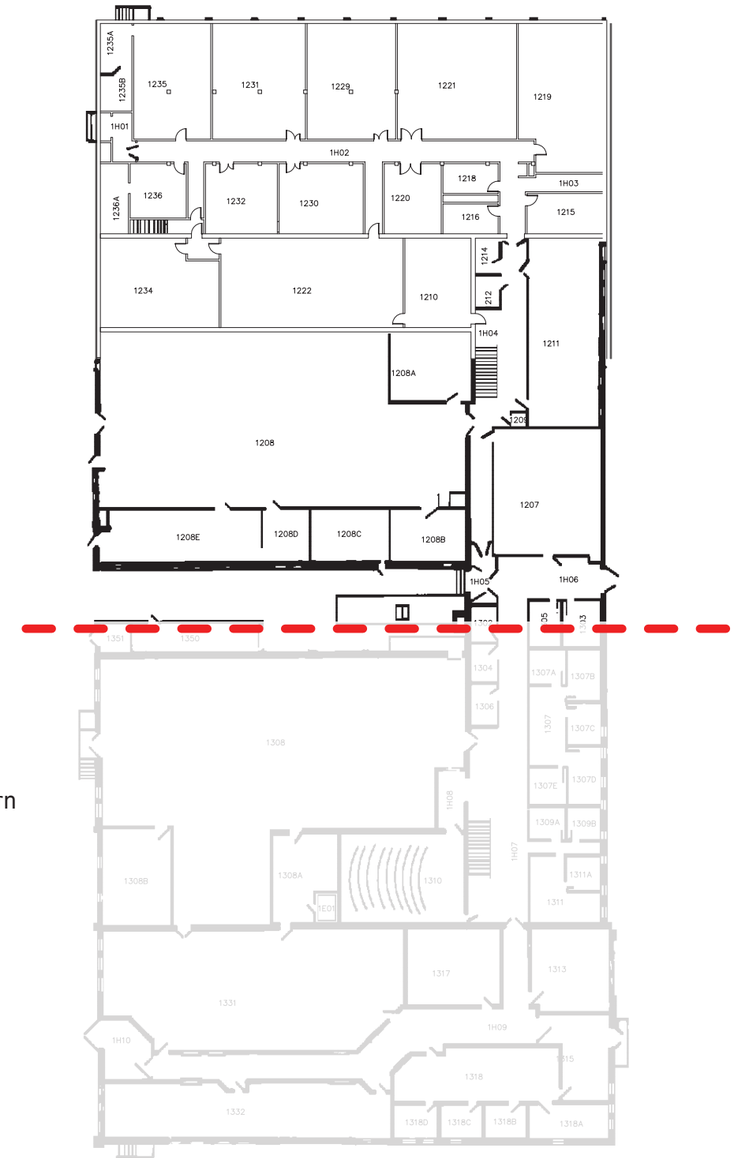
Loading entry

Landscaping zone

Pedestrian entry

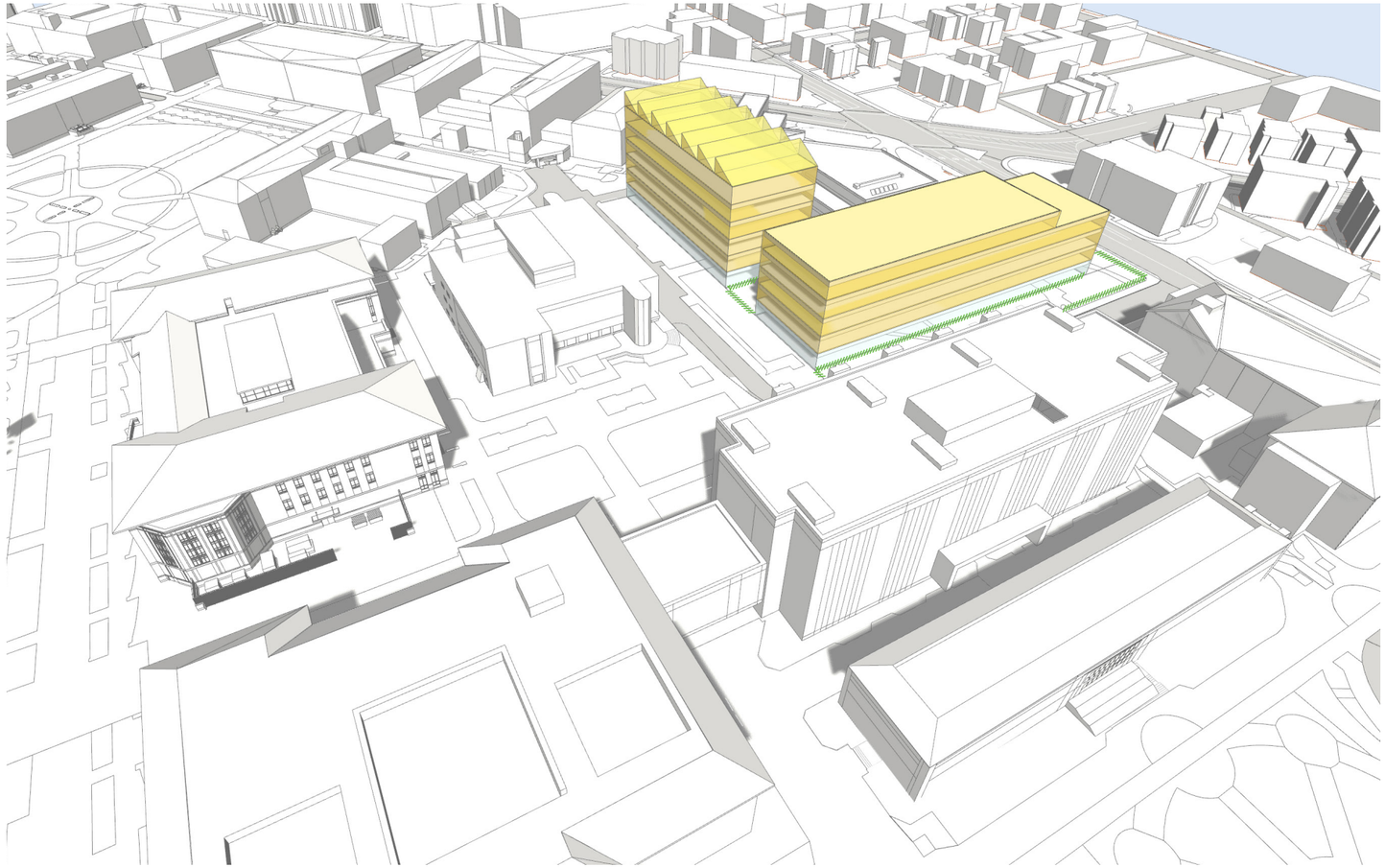
2. Context: Michael Golden Laboratories and Shops (MGL)

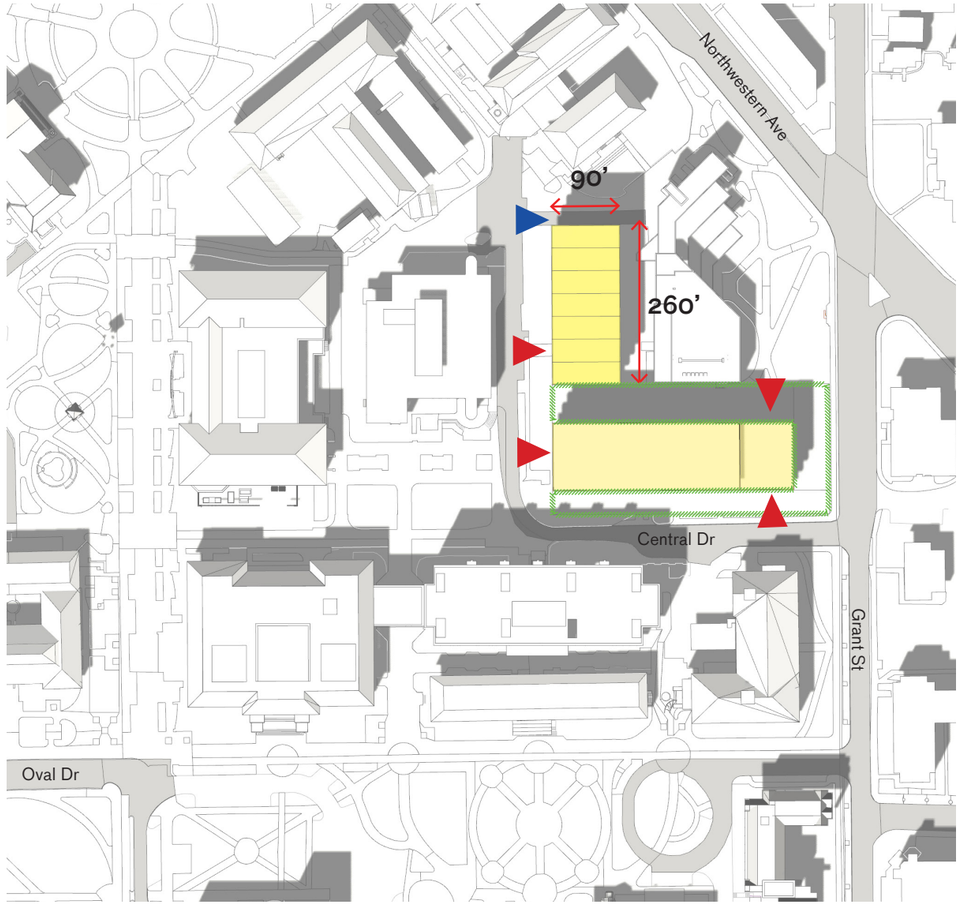




Level 1

Quad: Phase 2





Floors: B+6
GSF: 132,202

	Phase 1	Phase 2
B	28,575	18,886
1	28,575	18,886
2	28,575	18,886
3	28,575	18,886
4	28,575	18,886
5	22,208	18,886
6		18,886
Totals	165,083	132,202
		297,285

1 Complete demolition of MGL

2 Established shared loading dock entry for knoy and phases 1 and 2

3 New Landscaping between Knoy and Phase 1

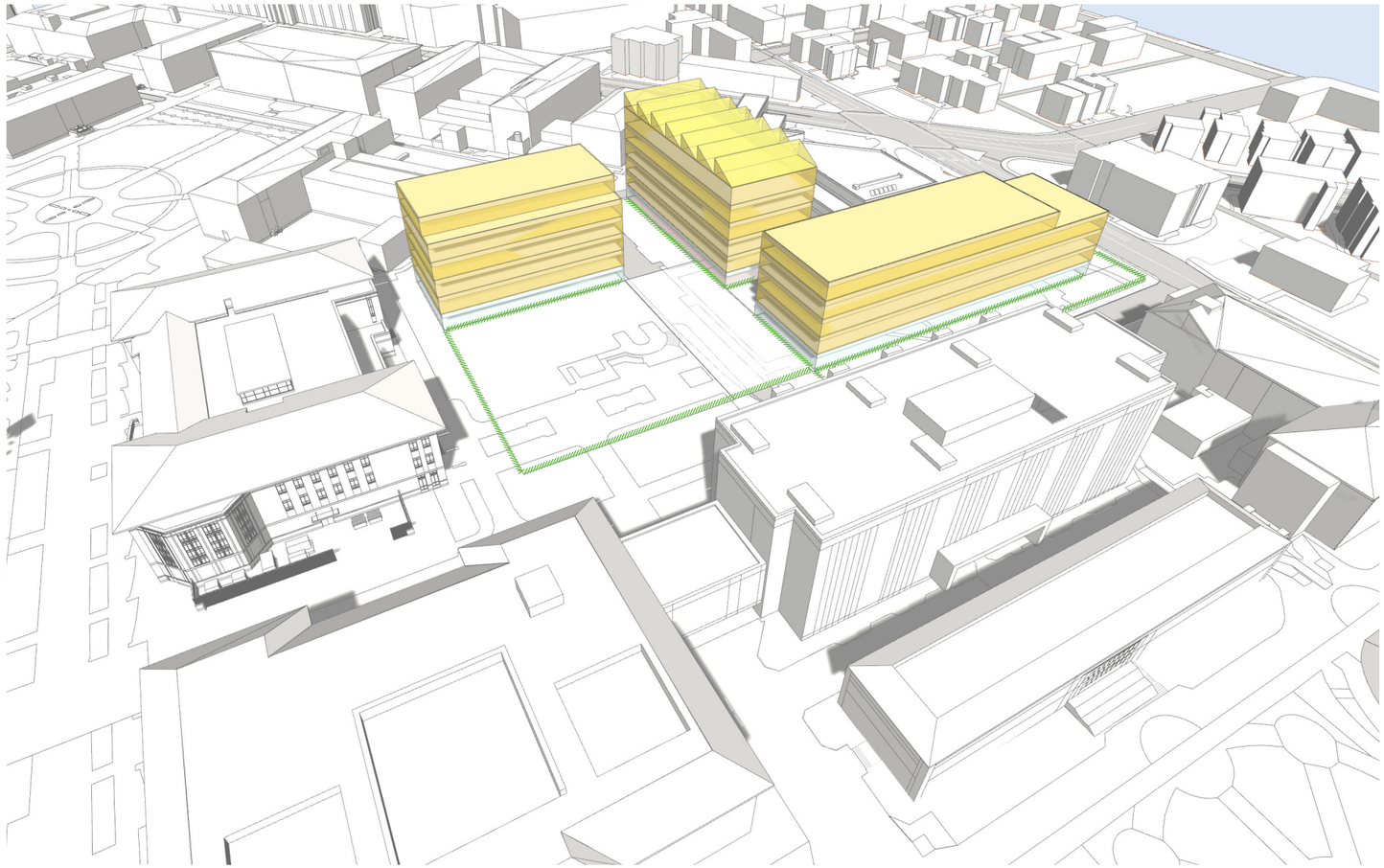
Loading dock

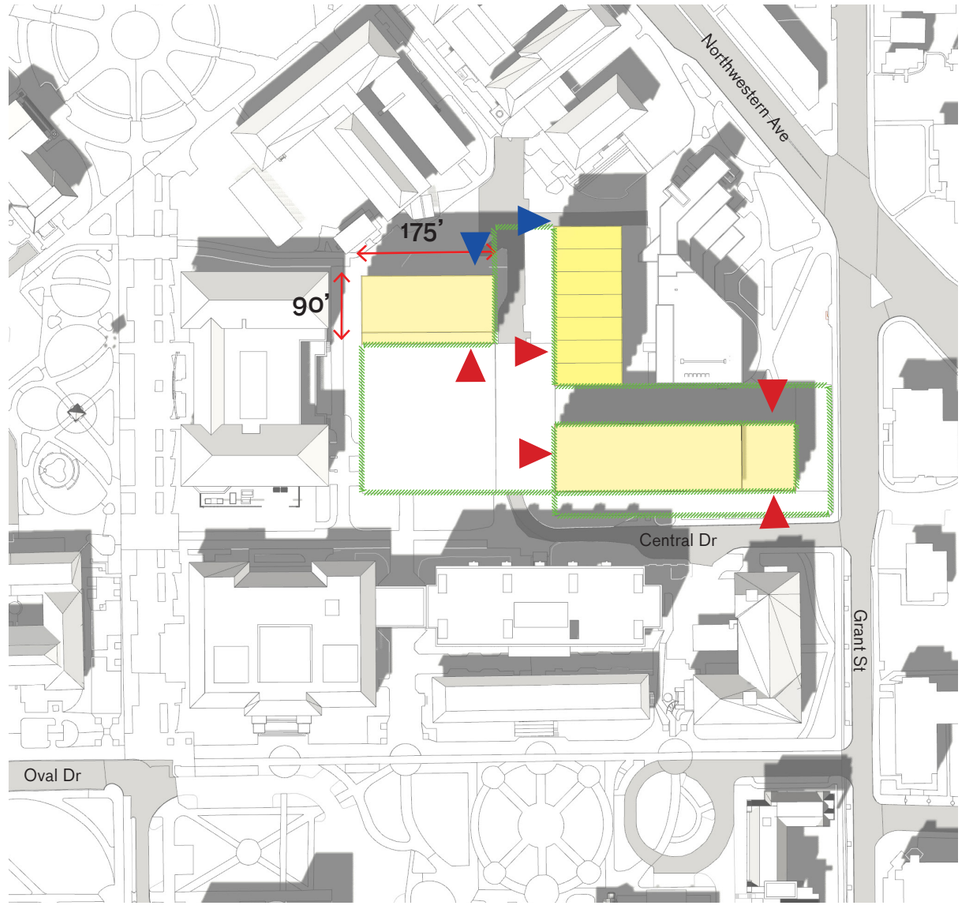
Loading entry

Landscaping zone

Pedestrian entry

Quad: Phase 3





1 Demolish Potter

2 New Loading Dock for phase 2 and 3 shared from new north entry

Floors:	**B+6**
GSF:	**107,625**

	Phase 1	Phase 2	Phase 3
B	28,575	18,886	15,750
1	28,575	18,886	15,750
2	28,575	18,886	15,750
3	28,575	18,886	15,750
4	28,575	18,886	15,750
5	22,208	18,886	15,750
6		18,886	13,125
Totals	**165,083**	**132,202**	**107,625**
			404,910

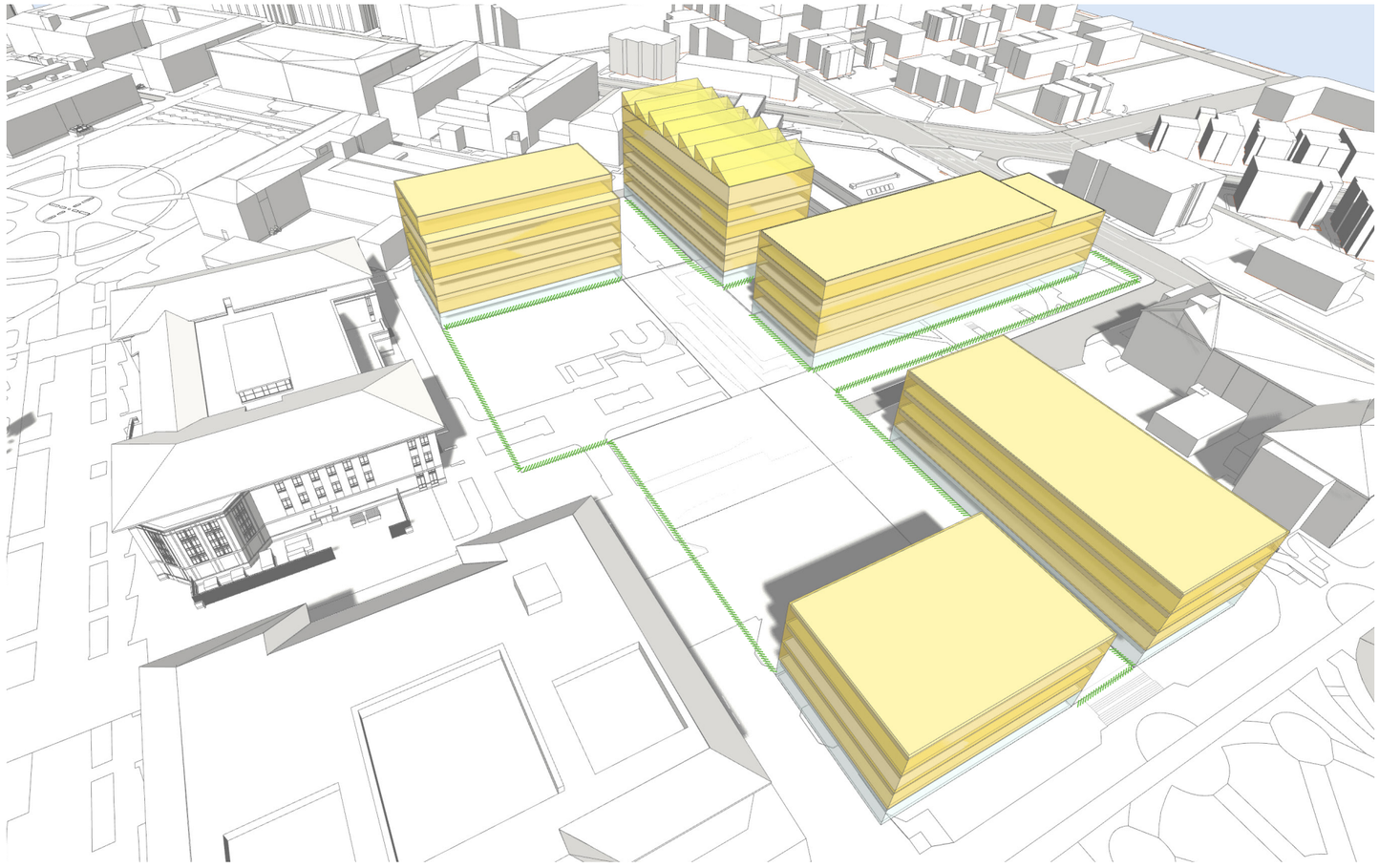
Loading dock

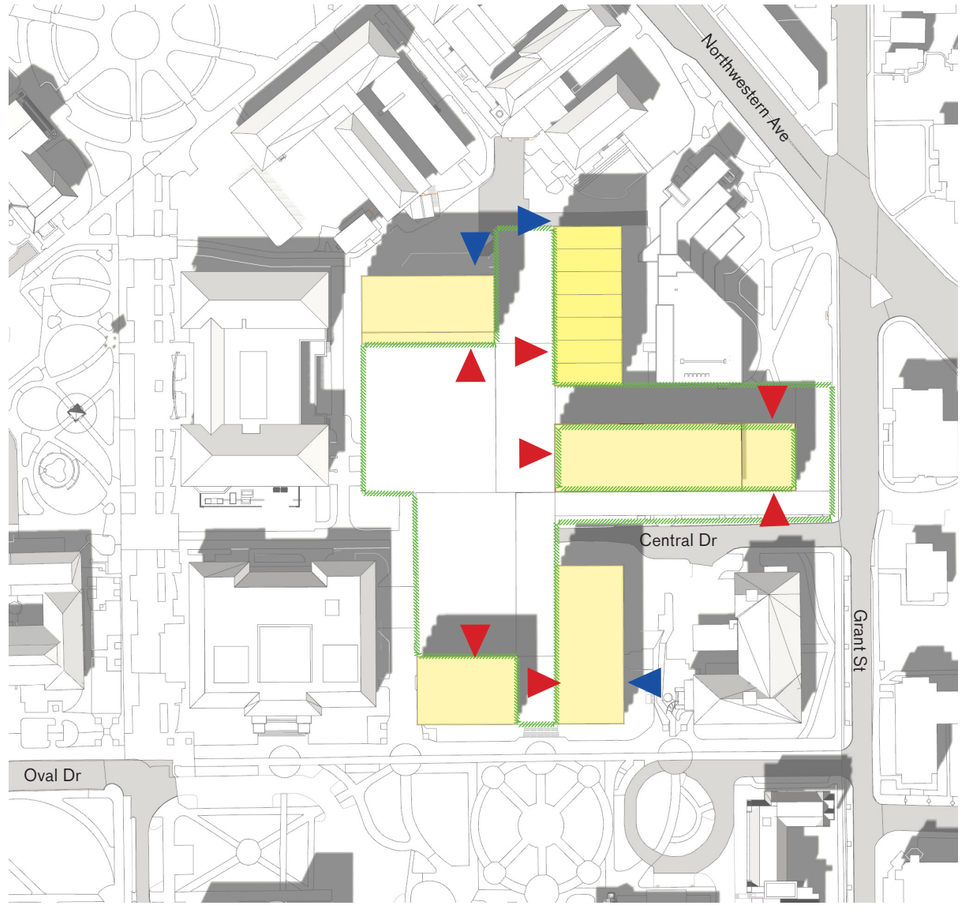
Loading entry

Landscaping zone

Pedestrian entry

Quad: Phase 4





1 Demolish Brown and Heavilon

2 Shared Surface extended across Central Drive

The Quad

	Phase 1	Phase 2	Phase 3
B	28,575	18,886	15,750
1	28,575	18,886	15,750
2	28,575	18,886	15,750
3	28,575	18,886	15,750
4	28,575	18,886	15,750
5	22,208	18,886	15,750
6		18,886	13,125
Totals	**165,083**	**132,202**	**107,625**
		297,285	**404,910**

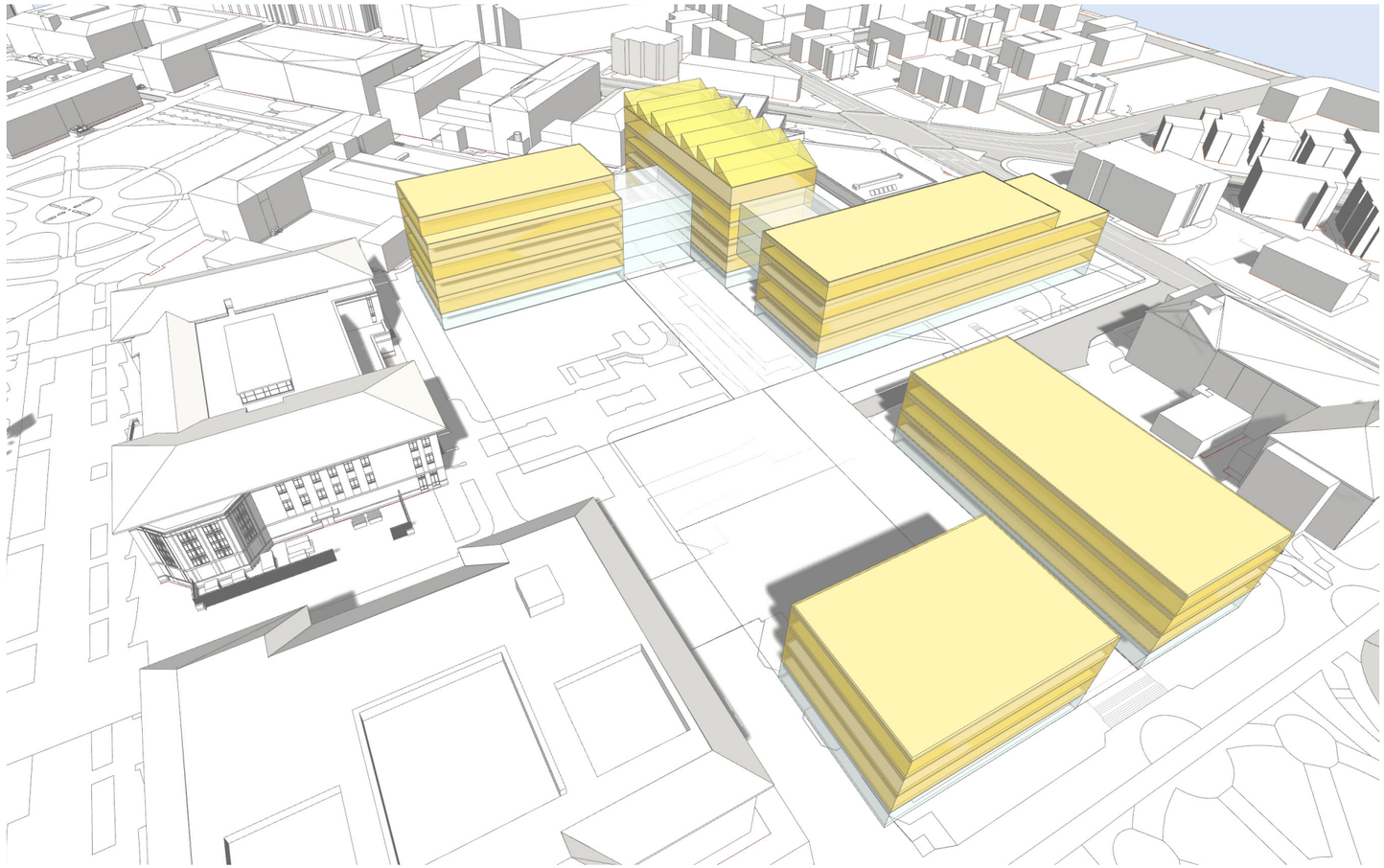
Loading dock

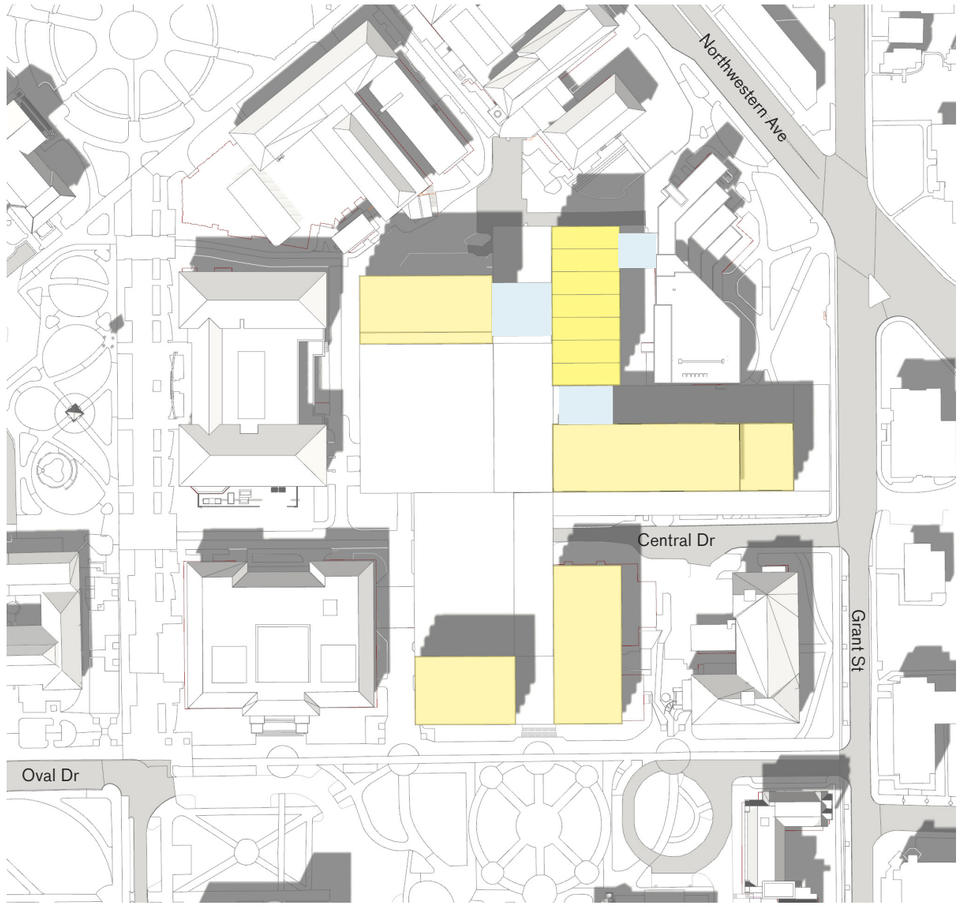
Loading entry

Landscaping zone

Pedestrian entry

Quad: Glass Bridges and Atria

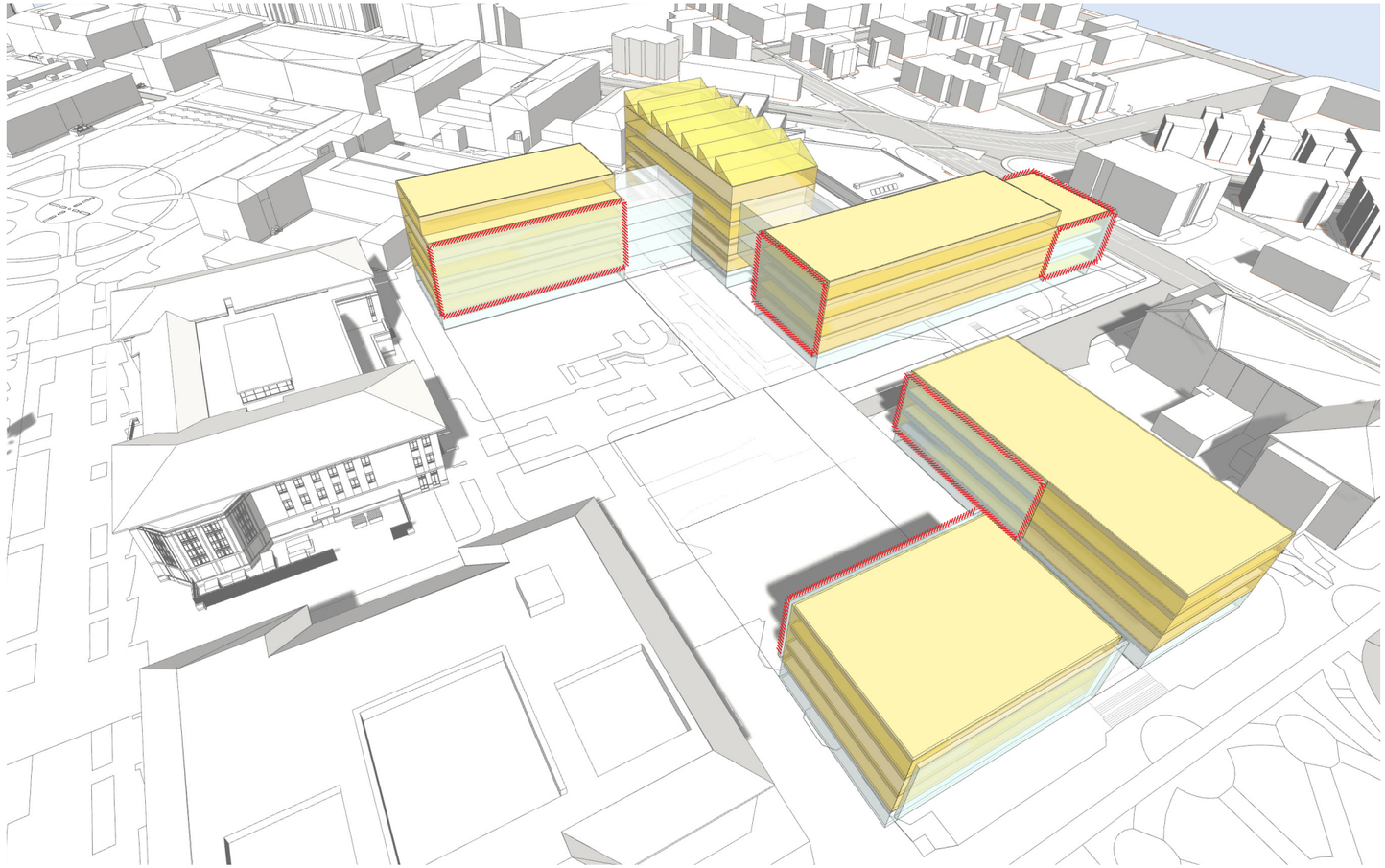


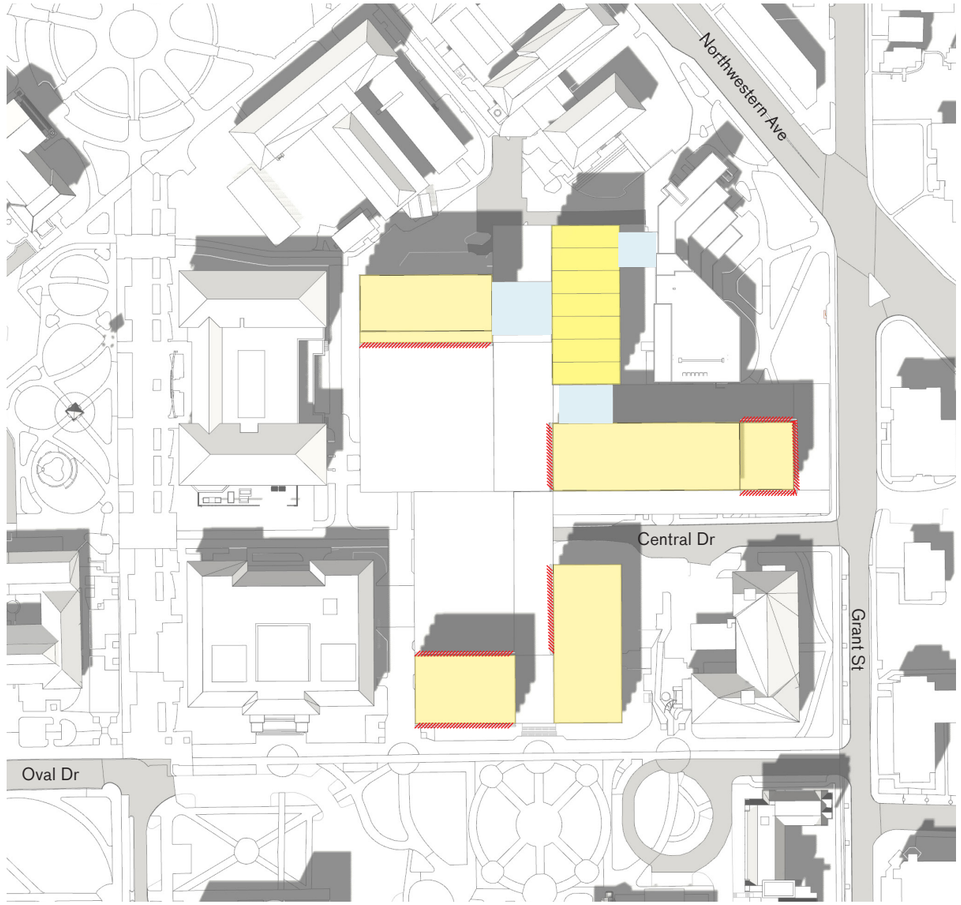


The Quad

	Phase 1	Phase 2	Phase 3	Bridge between phase 1 & 2	Bridge between phase 2 & Knov	Atrium between phases 2 & 3	
B	28,575	18,886	15,750	751		1,167	
1	28,575	18,886	15,750			5,447	
2	28,575	18,886	15,750			5,447	
3	28,575	18,886	15,750	3,488		5,447	
4	28,575	18,886	15,750	3,488	2,305	5,447	
5	22,208	18,886	15,750	3,488	2,305		
6		18,886	13,125				
							GRAND TOTAL
Totals	**165,083**	**132,202**	**107,625**	**11,215**	**4,610**	**22,955**	**443,690**
		297,285	**404,910**			**38,780**	

Quad: Active and Passive Facades

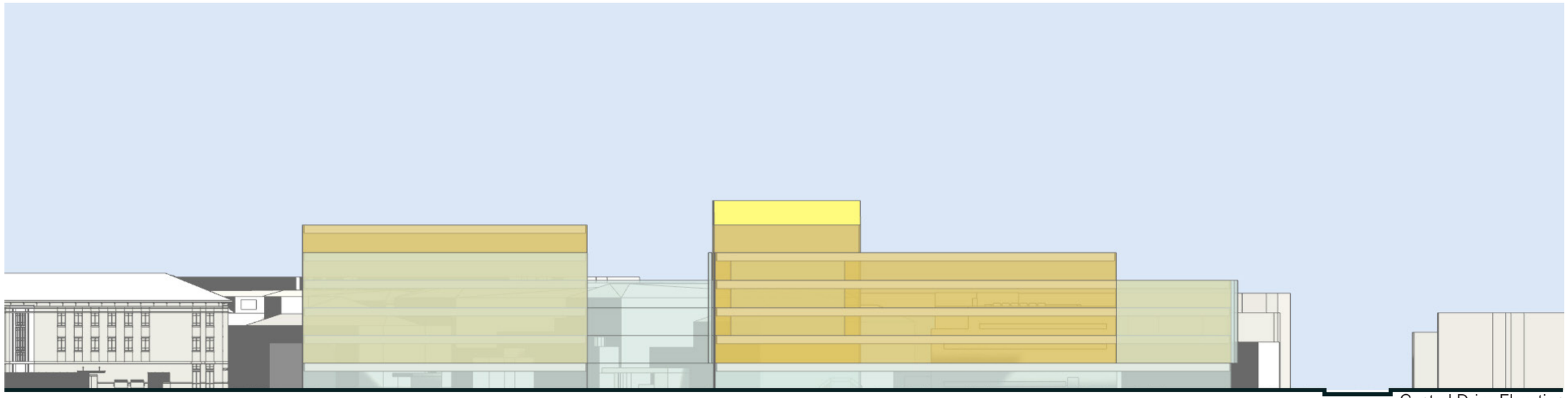




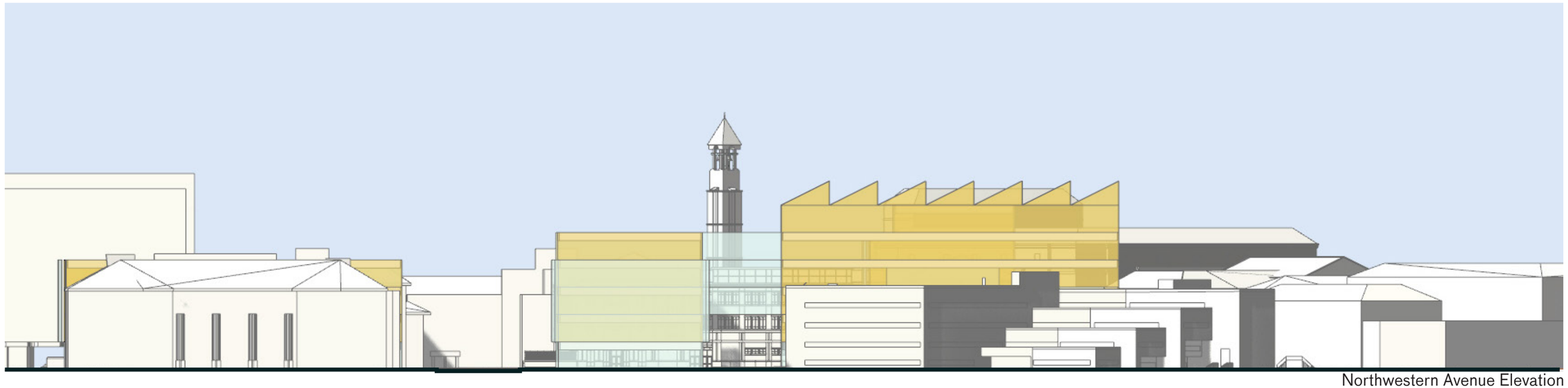
The Quad

	Phase 1	Phase 2	Phase 3	Bridge between phase 1 & 2	Bridge between phase 2 & Knov	Atrium between phases 2 & 3	
B	28,575	18,886	15,750	751		1,167	
1	28,575	18,886	15,750			5,447	
2	28,575	18,886	15,750			5,447	
3	28,575	18,886	15,750	3,488		5,447	
4	28,575	18,886	15,750	3,488	2,305	5,447	
5	22,208	18,886	15,750	3,488	2,305		
6		18,886	13,125				
							GRAND TOTAL
Totals	**165,083**	**132,202**	**107,625**	**11,215**	**4,610**	**22,955**	**443,690**
		297,285	**404,910**			**38,780**	

Quad: Elevations

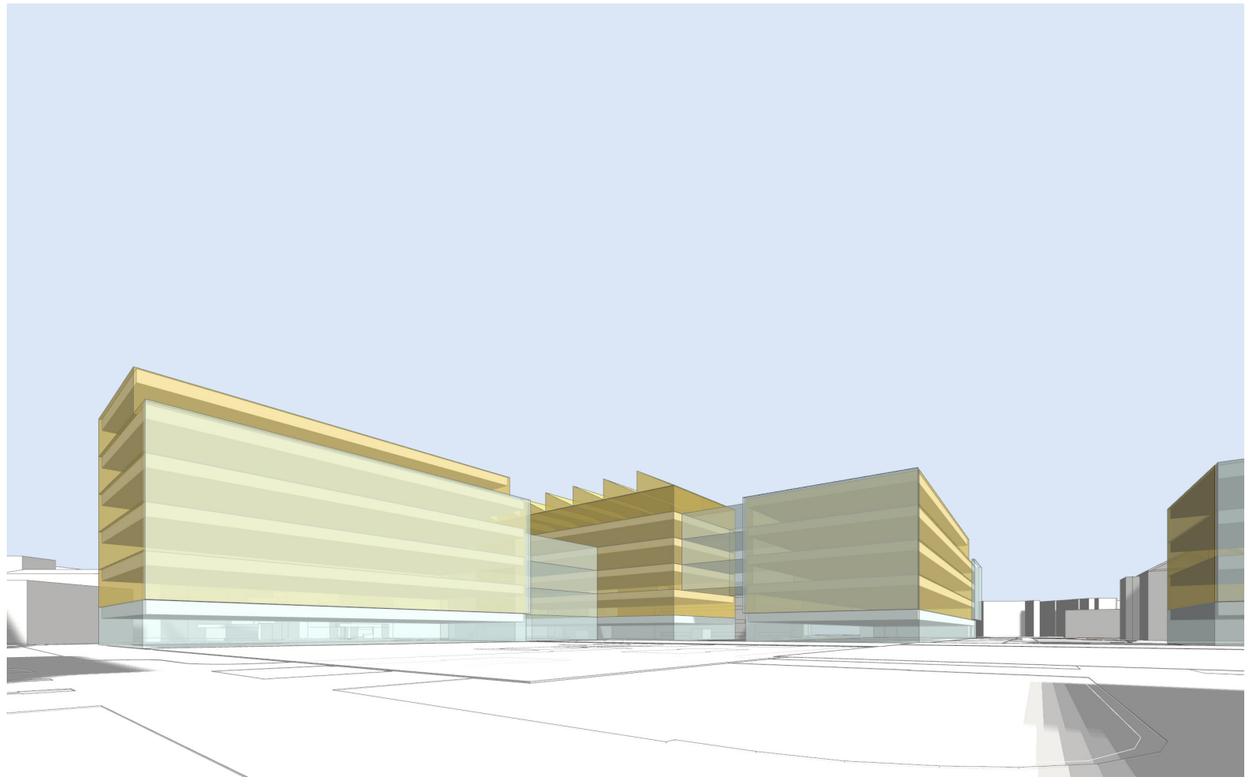


Central Drive Elevation

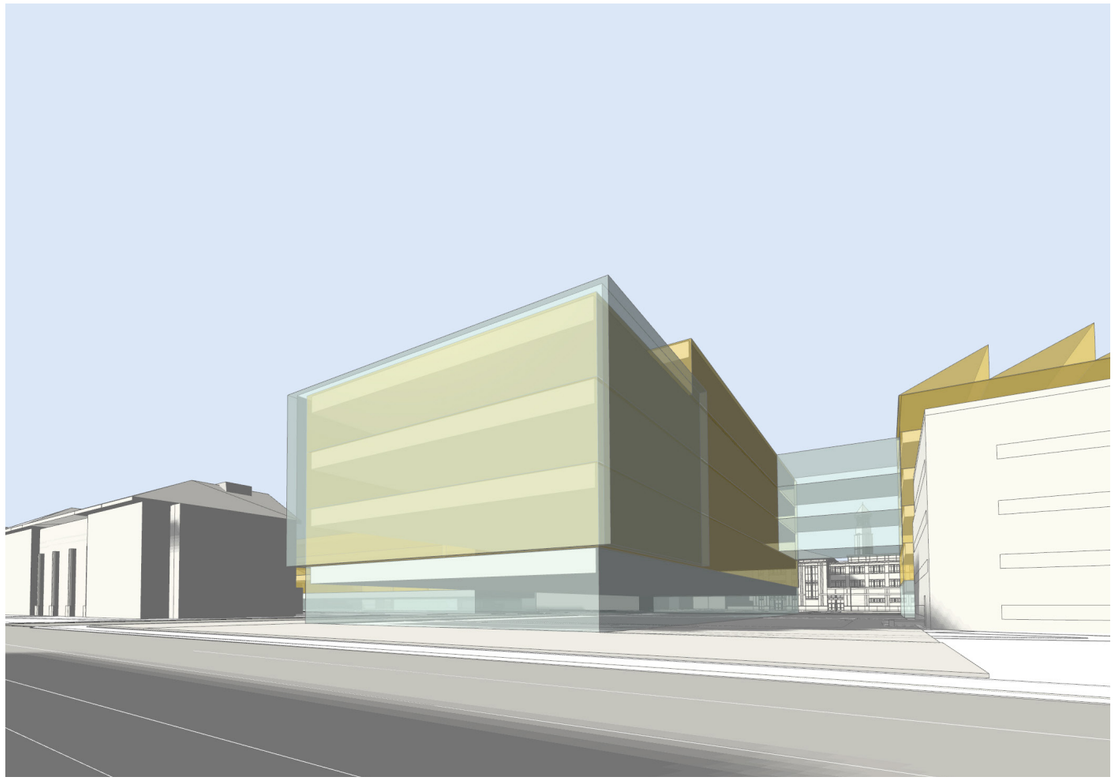


Northwestern Avenue Elevation

Quad: Street level views



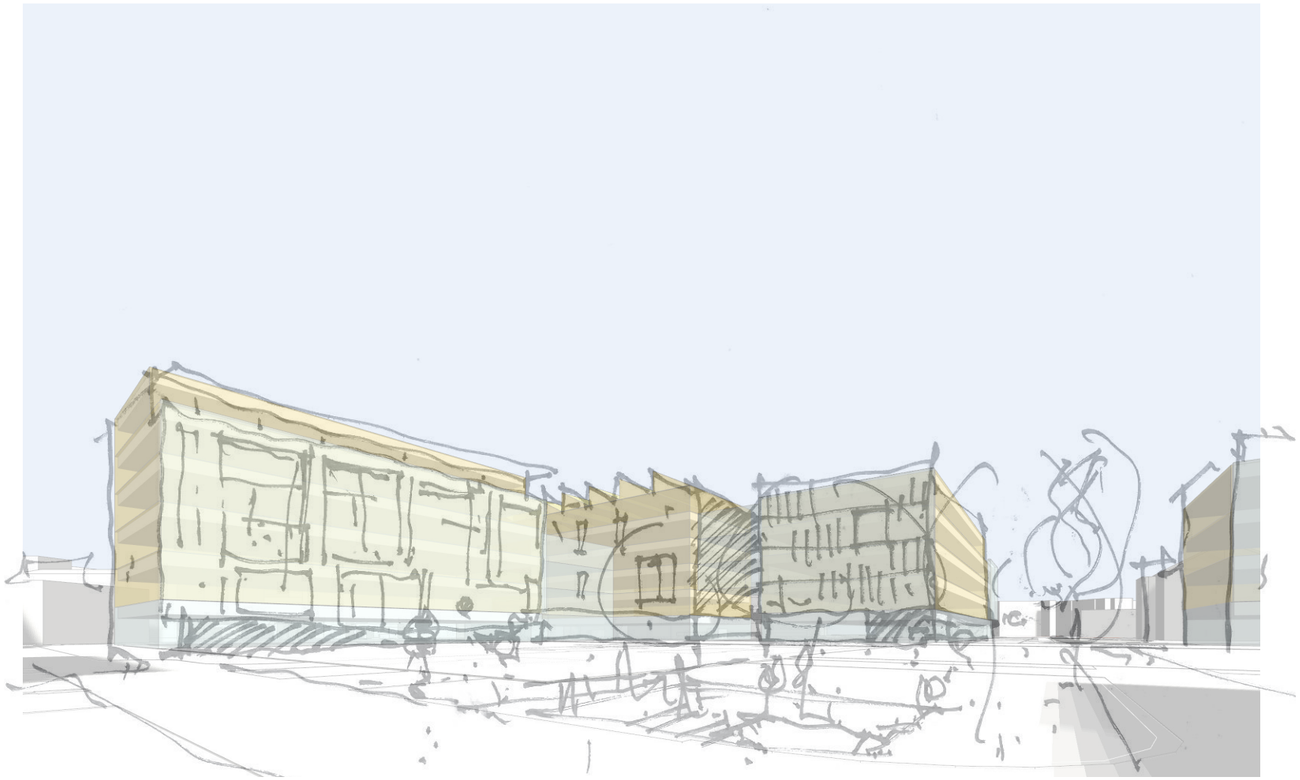
Quad: Courtyard perspective, final build-out



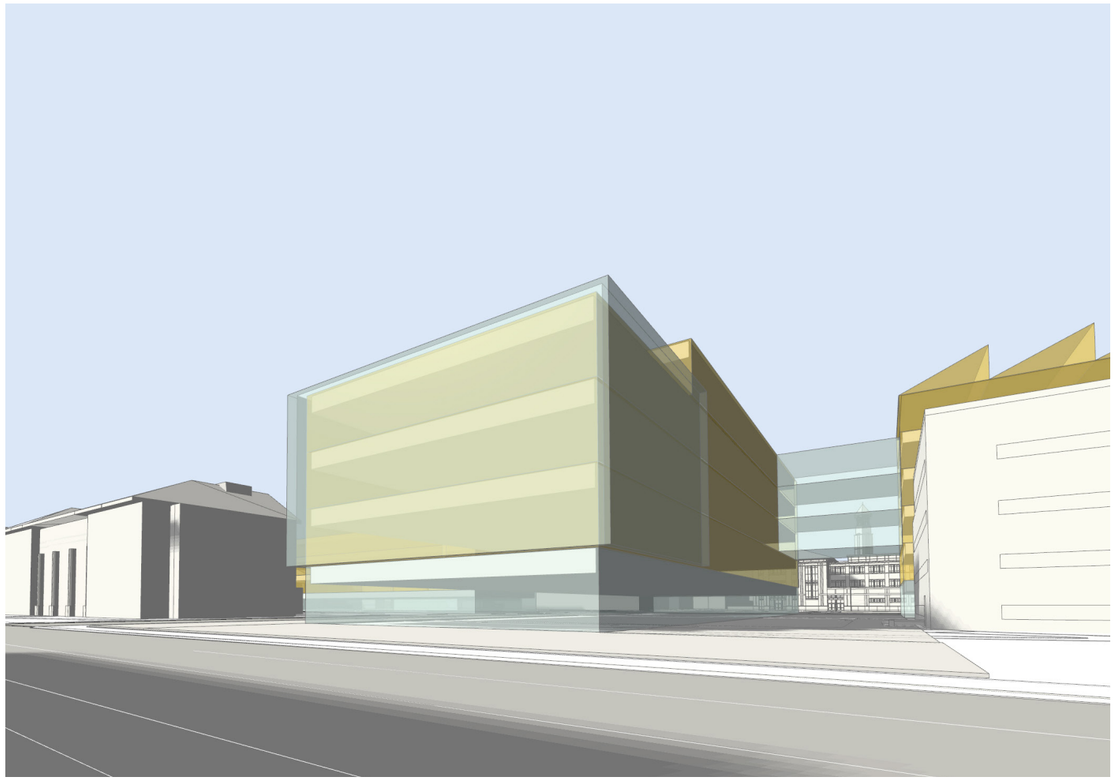
Quad: Northwestern Avenue perspective

Quad: Courtyard perspective, Phase 1

Quad: Street level views



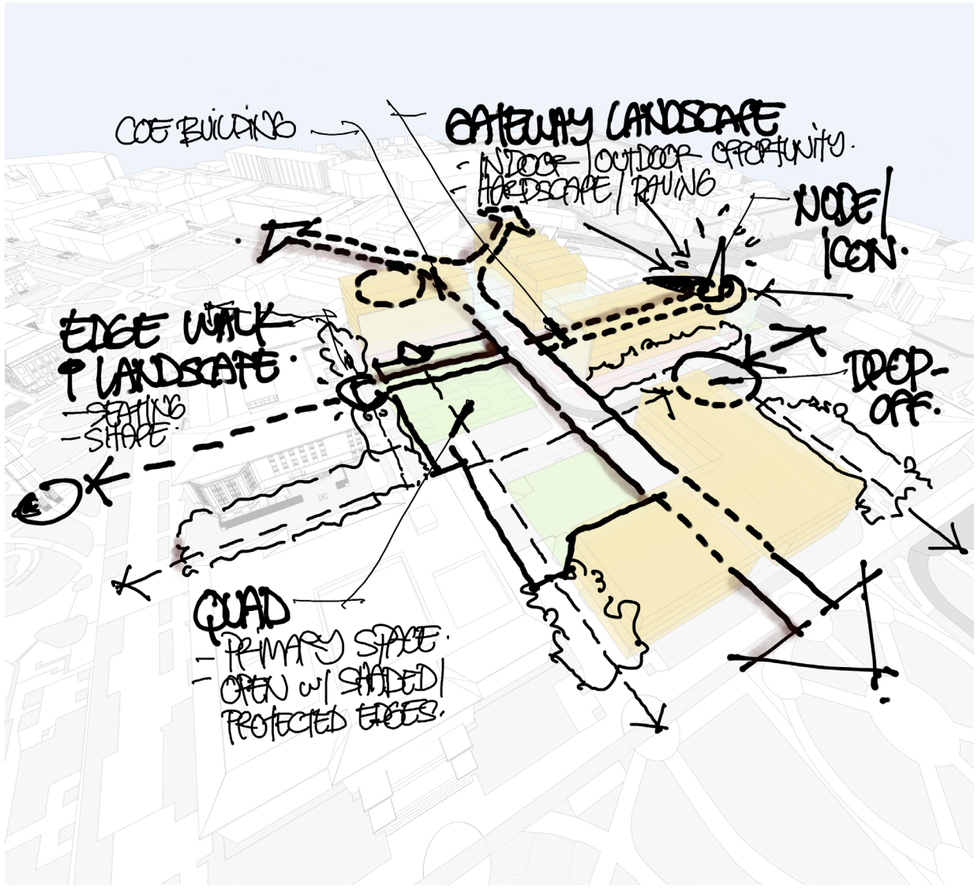
Quad: Courtyard perspective, final build-out

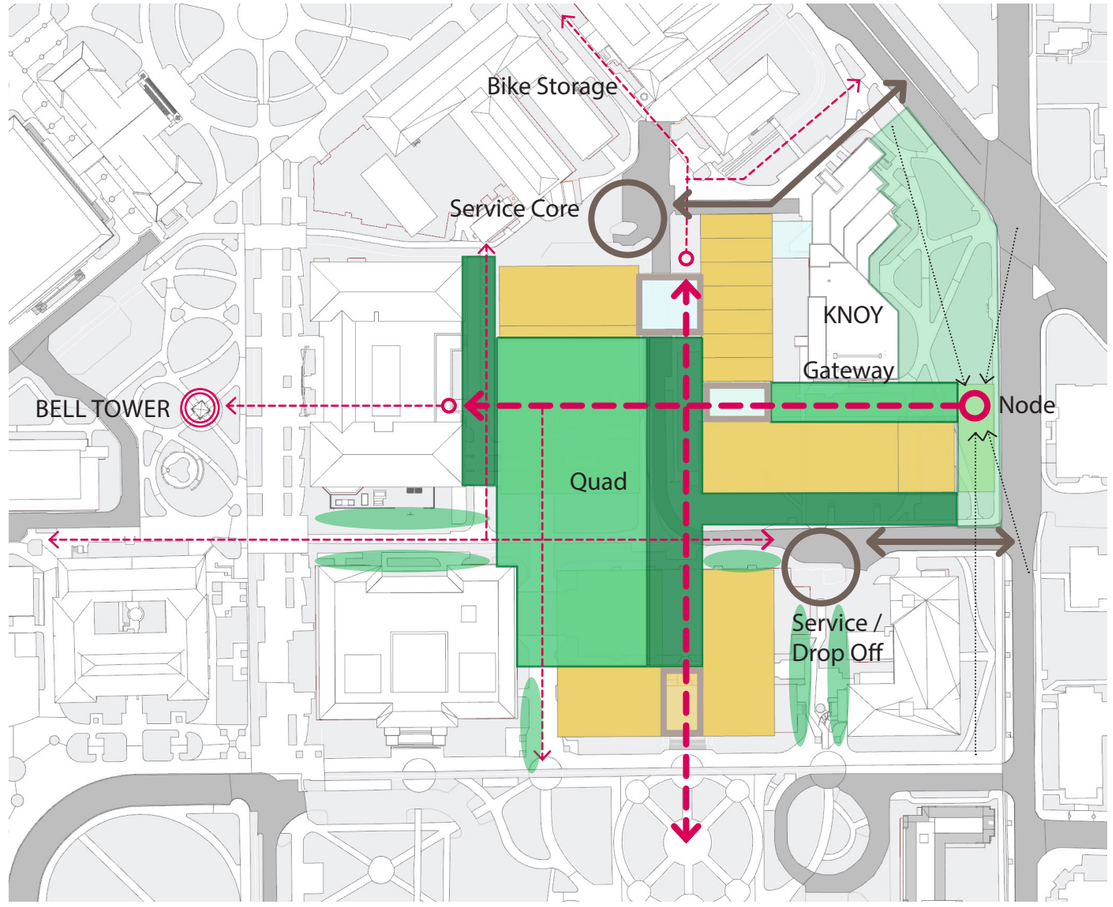


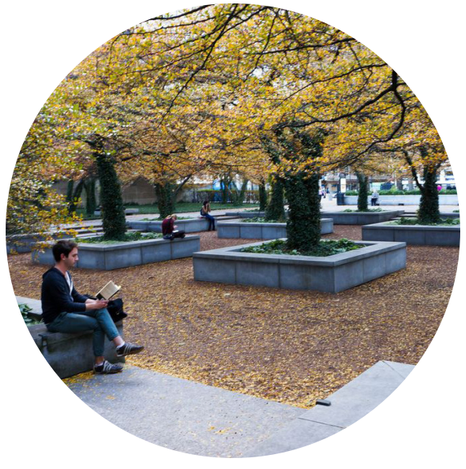
Quad: Northwestern Avenue perspective

Quad: Courtyard perspective, Phase 1

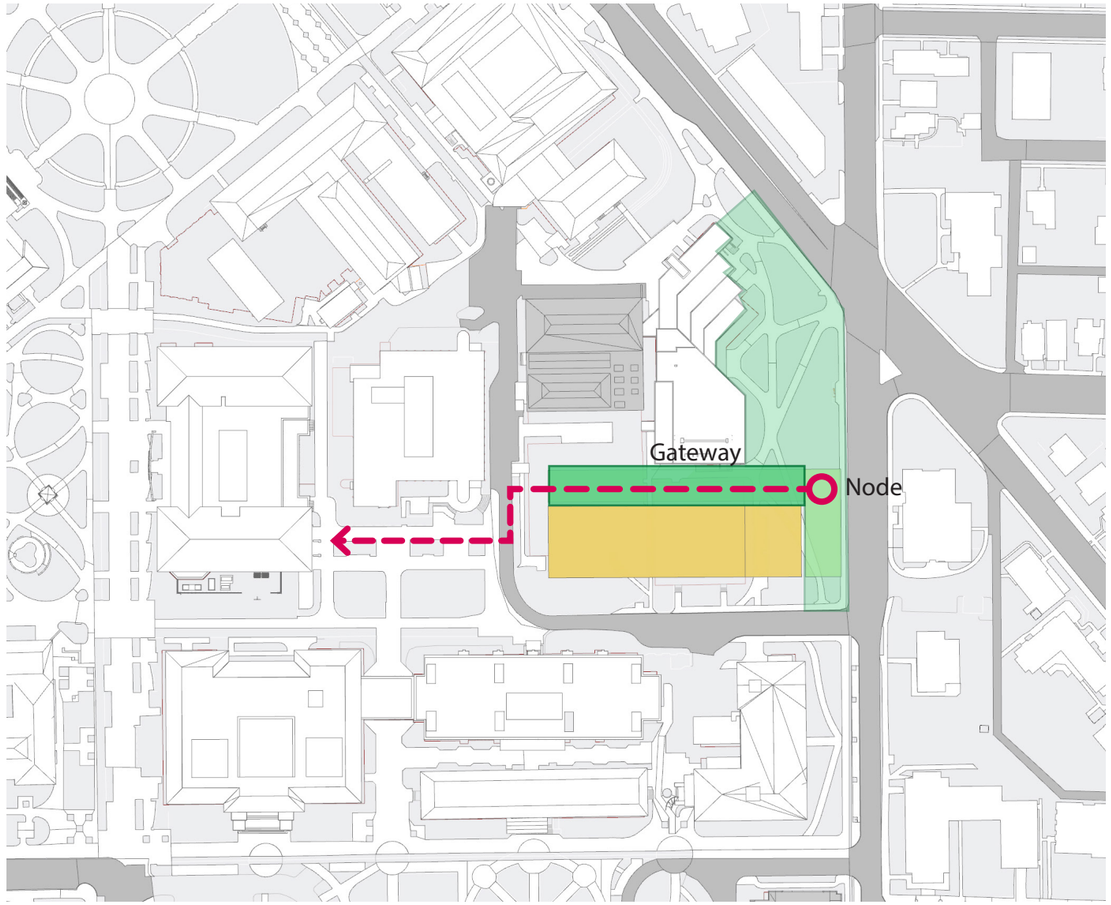
Quad: Landscape







Quad: Landscape Phasing Approach

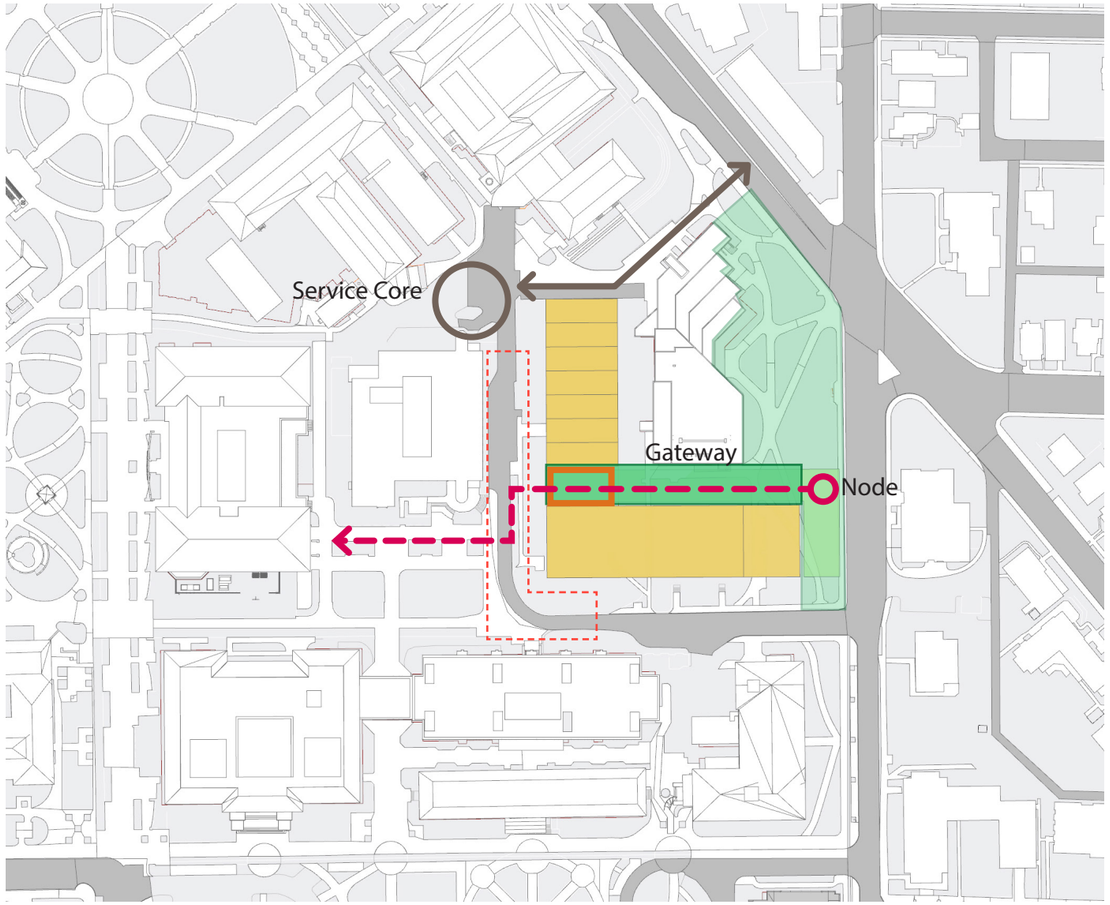


Phase 1

1 Establish gateway landscape between CoE and Knoy, and make connection to ALC

2 Announce gateway with landscape node or feature - artwork, curated student work, water, color, light

3 Revise Knoy landscape to support CoE gateway

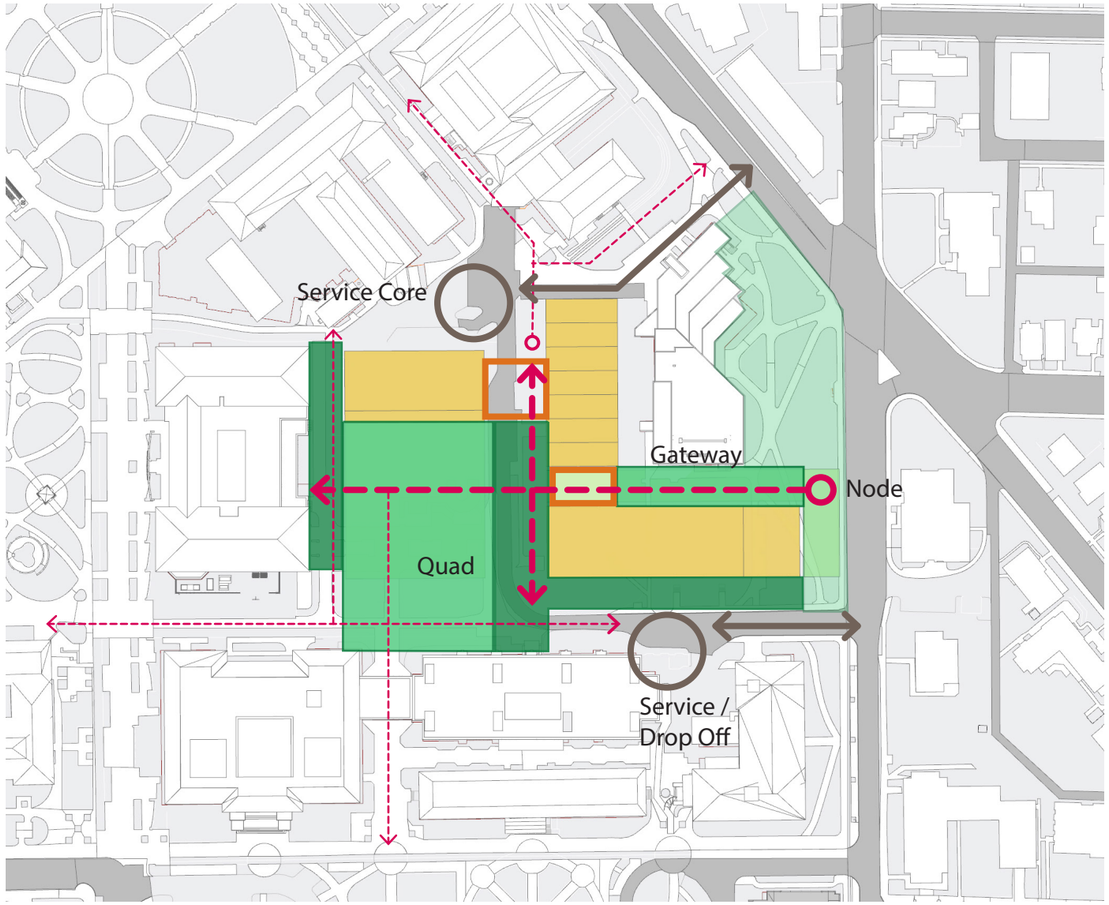


Phase 2

1 Create access to new Service Core, separate pedestrian/bike and vehicle pathways

2 Demolish existing vehicular roadways along western edge of CoE

Quad: Landscape Phasing Approach



Phase 3

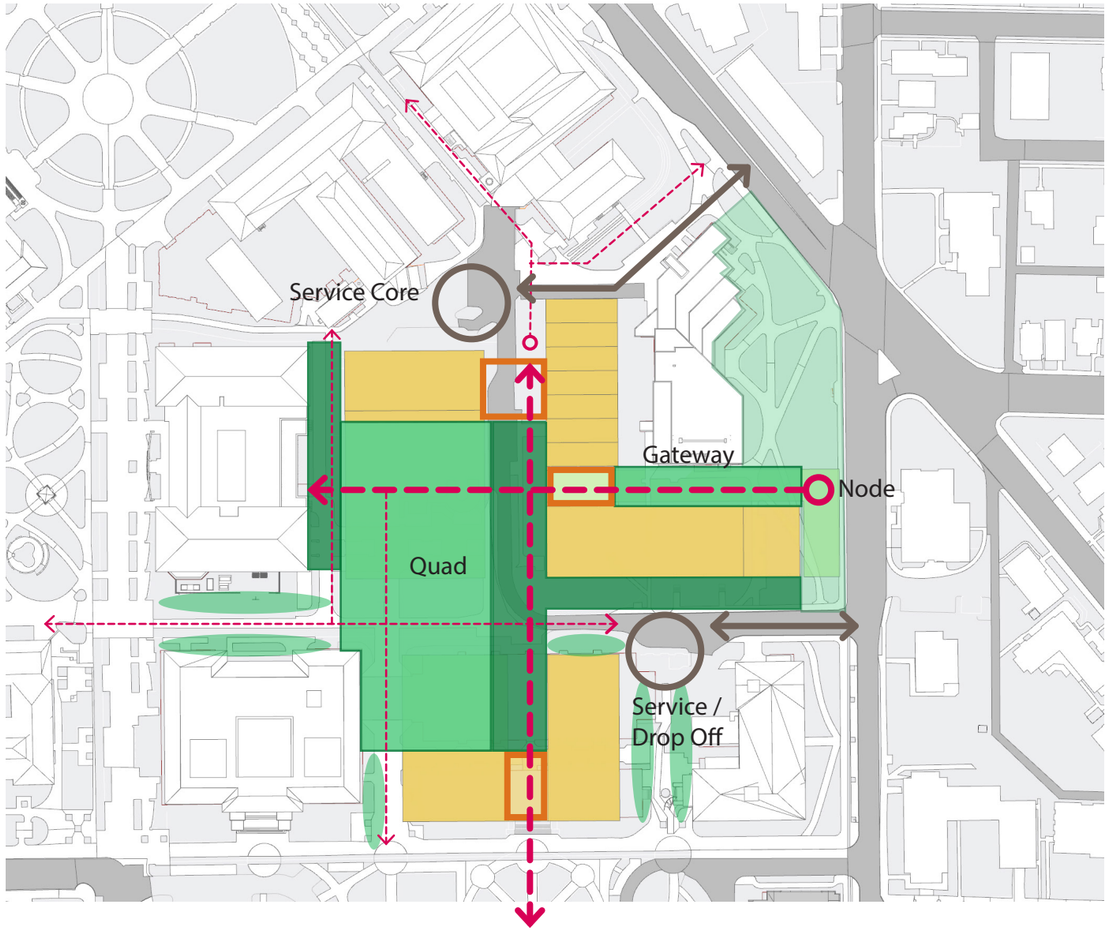
1 Reconstruct eastern Central Drive into a service access and potential drop-off area

2 Establish western Central Drive as pedestrian connection

3 Straighten Gateway / Promenade to align with main entry at ALC

4 Create N-S edge landscape and walk along CoE; connect to bike/ped connection to the north

5 Establish northern section of quad, along with ALC edge landscape



Phase 4

1 Complete quad landscape and extend edge landscapes to the south

2 Enhance tree canopy in select areas along primary walks

4. Massing Option 2: The Serpentine

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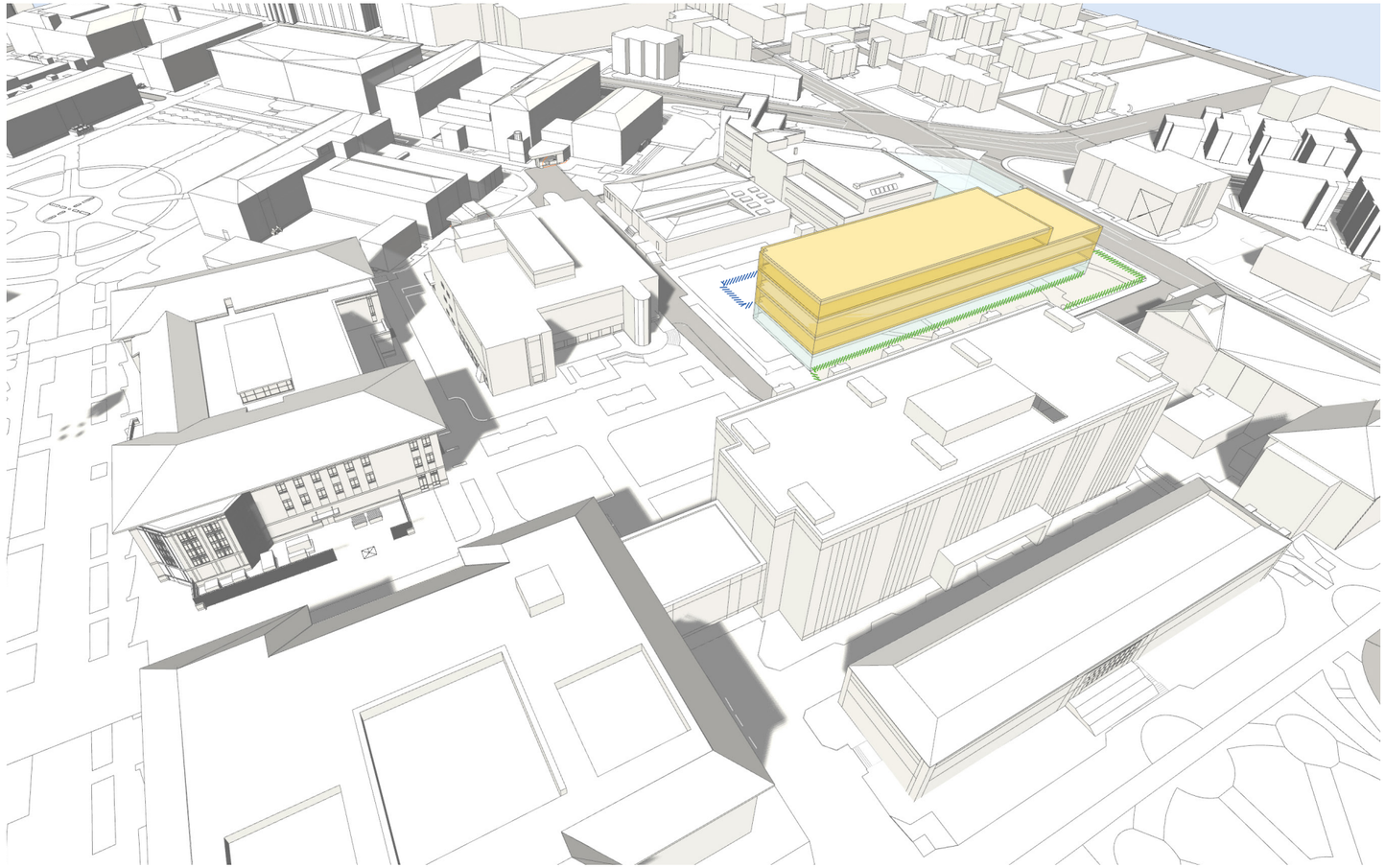
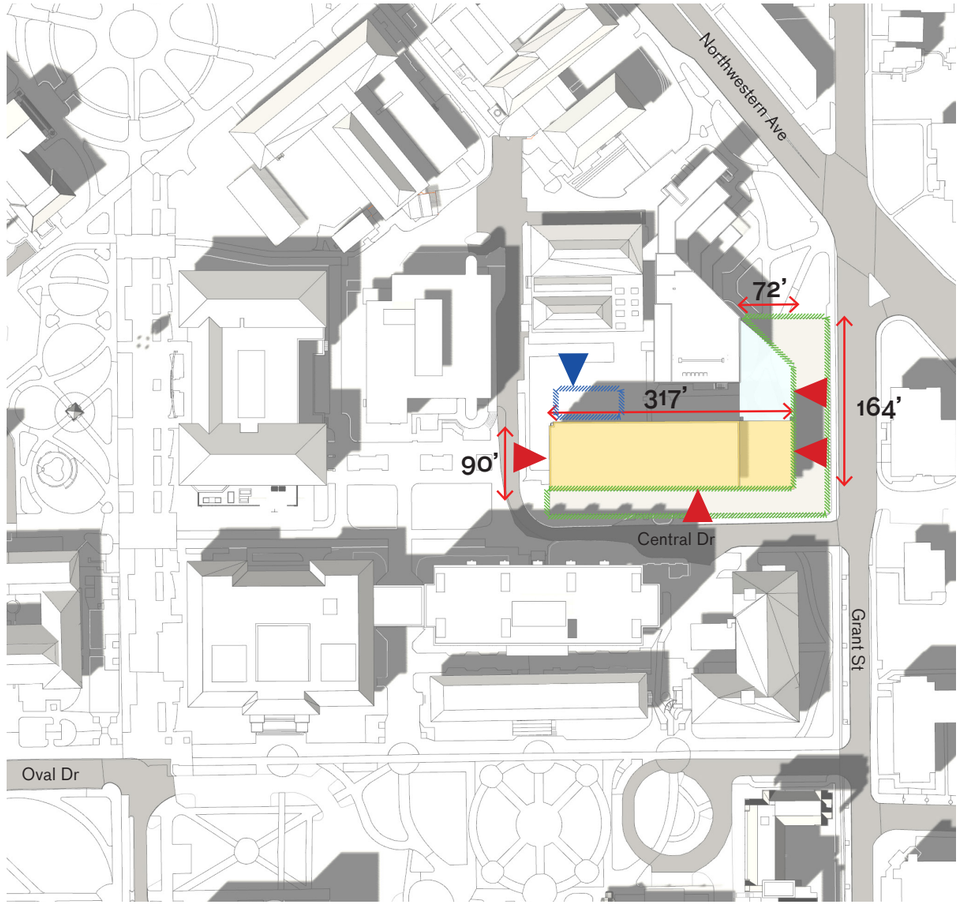
Serpentine: Phase 1





Floors:	**B+4**
GSF:	**168,885**

	Phase 1
B	36,591
1	36,591
2	36,591
3	36,591
4	22,521
5	
6	
Totals	**168,885**

1 Part demolition of MGL

2 Temporary Loading Established off Central Drive

3 New landscaped area to the south and eas of Phase 1

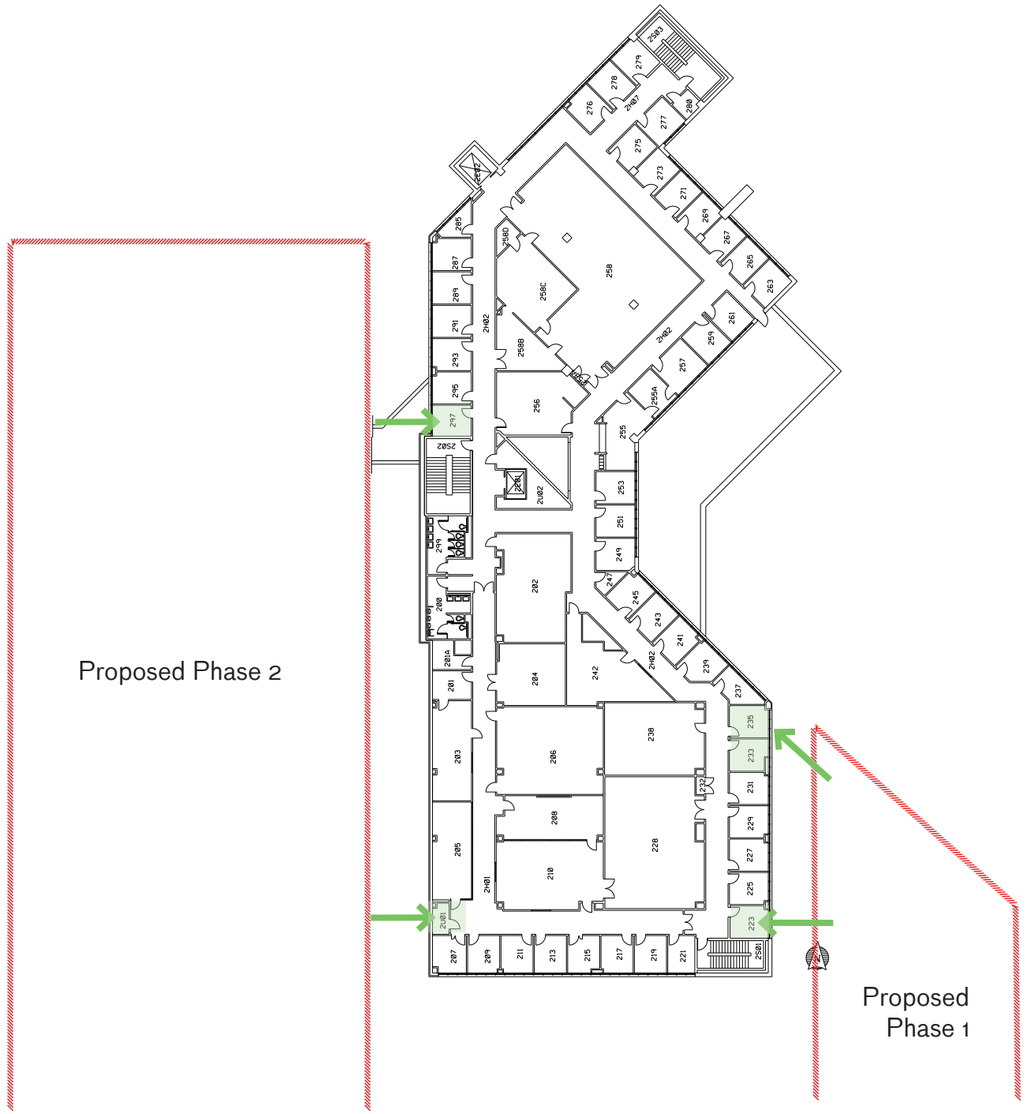
Loading dock

Loading entry

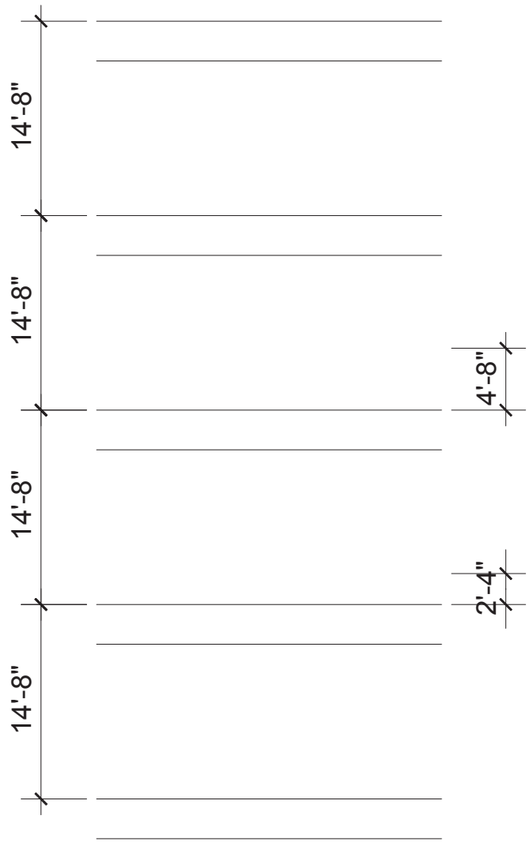
Landscaping zone

Pedestrian entry

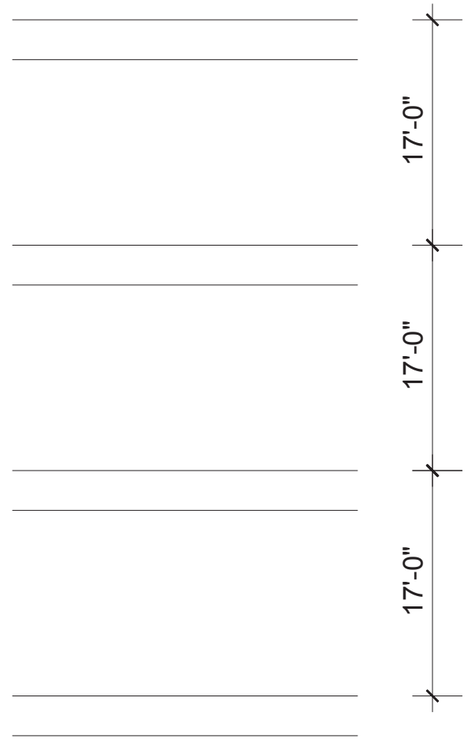
Serpentine: Knoy Hall Connection



Typical Plan Levels 2-4

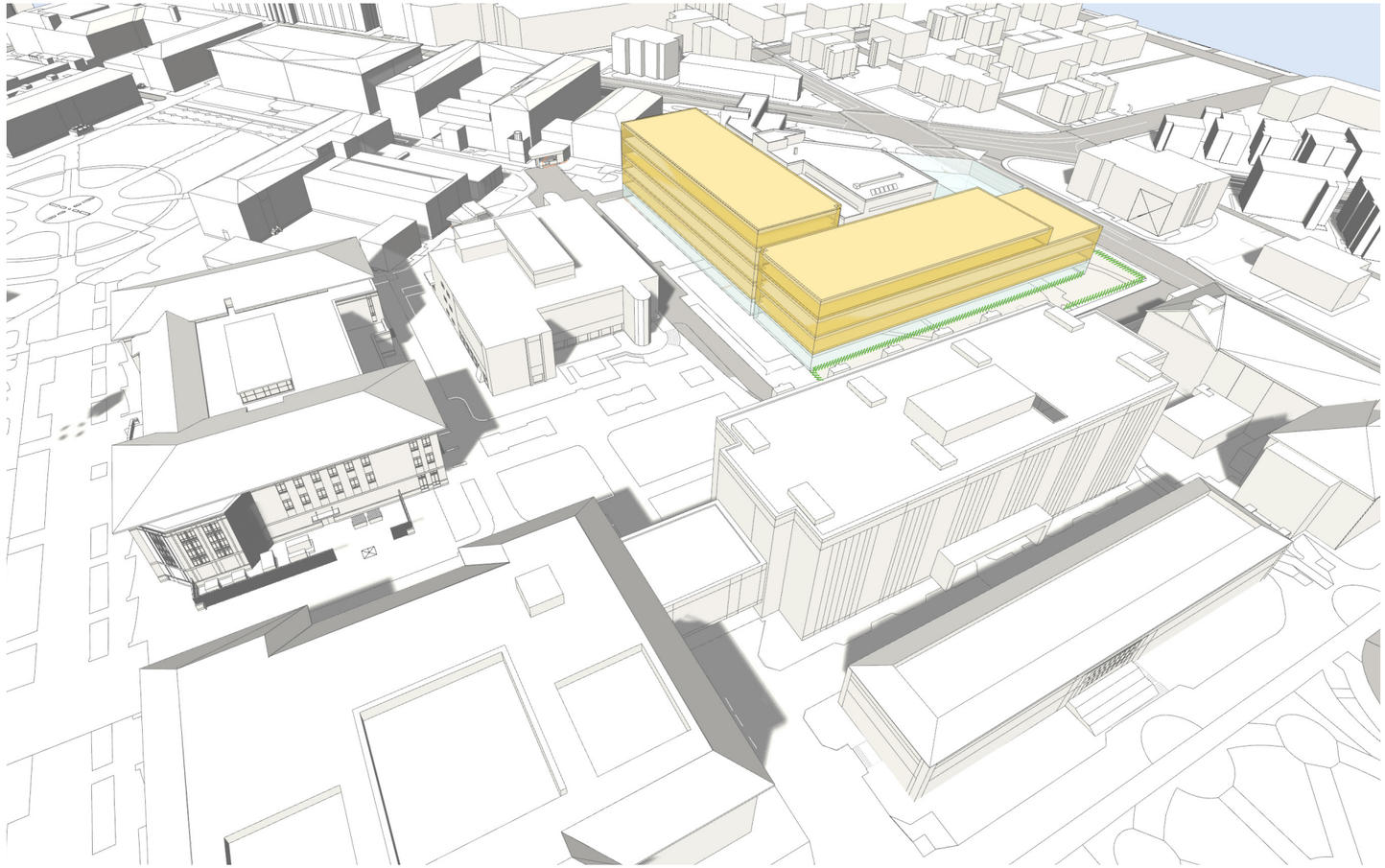


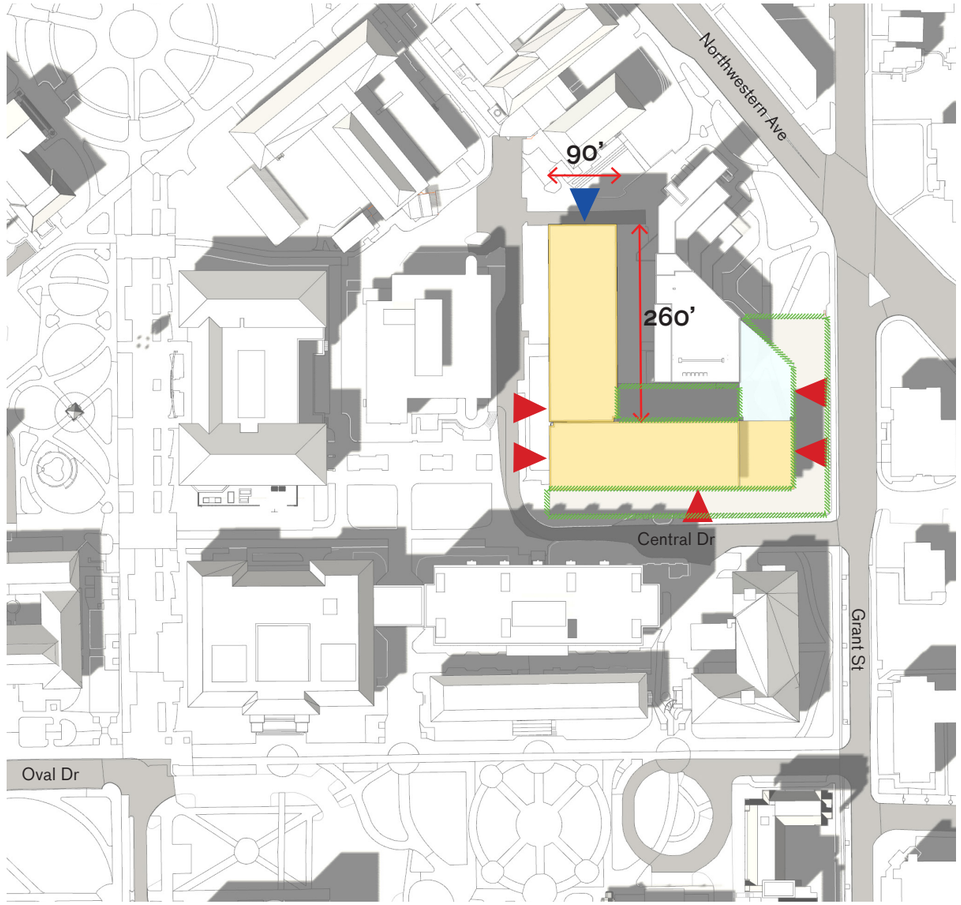
Existing Knoy Elevations
Levels 1-4 Floor-to-floor height: 14’-8”



Proposed Phase 1 and 2
Levels 1-3 floor-to-floor height: 17’-0”

Serpentine: Phase 2





Floors: B+5
GSF: 140,172

	Phase 1	Phase 2
B	36,591	23,362
1	36,591	23,362
2	36,591	23,362
3	36,591	23,362
4	22,521	23,362
5		23,362
6		
Totals	**168,885**	**140,172**

1 Complete demolition of MGL

2 Include shared loading dock entry for Knoy and phases 1 and 2 at north of phase 2

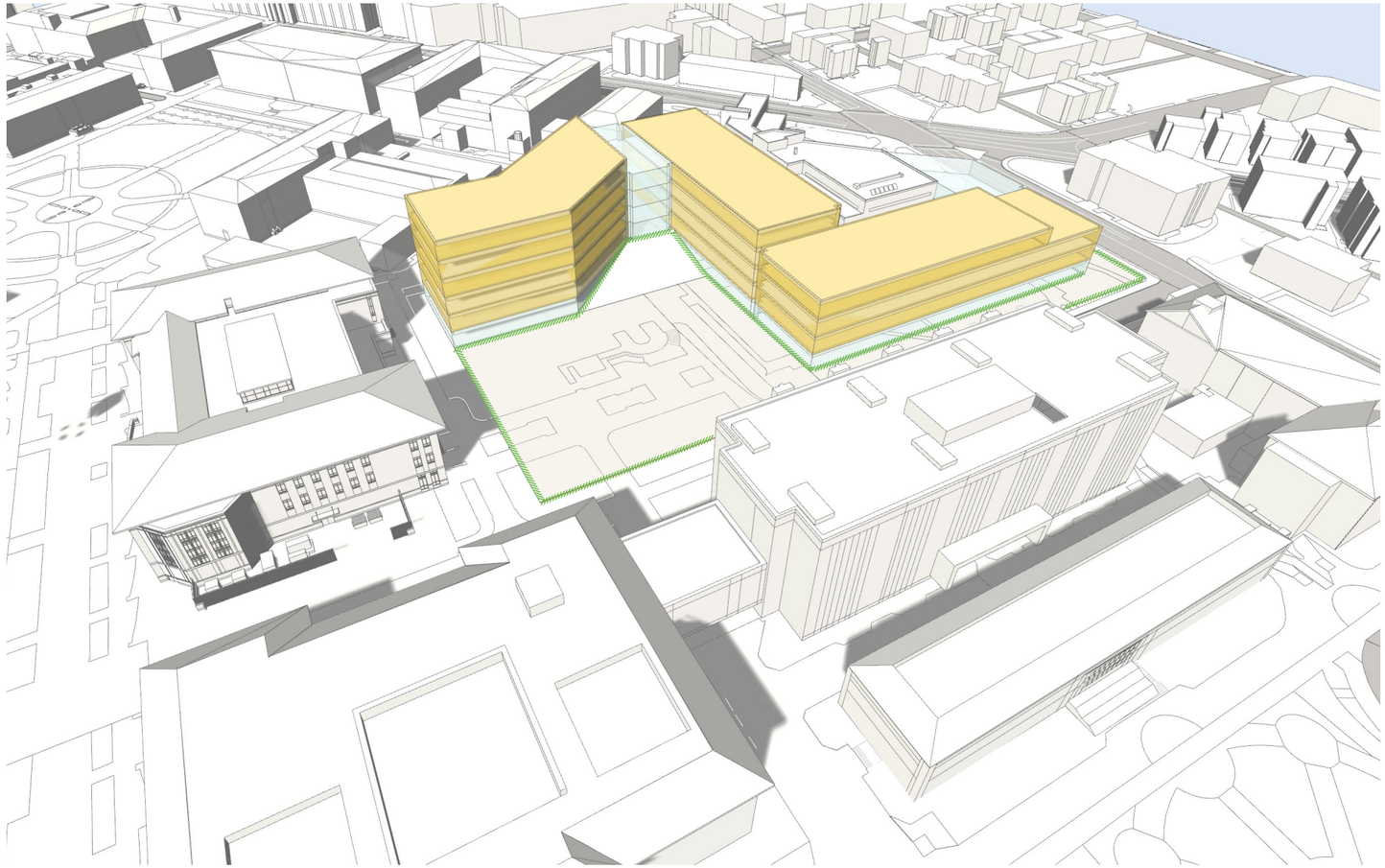
Loading dock

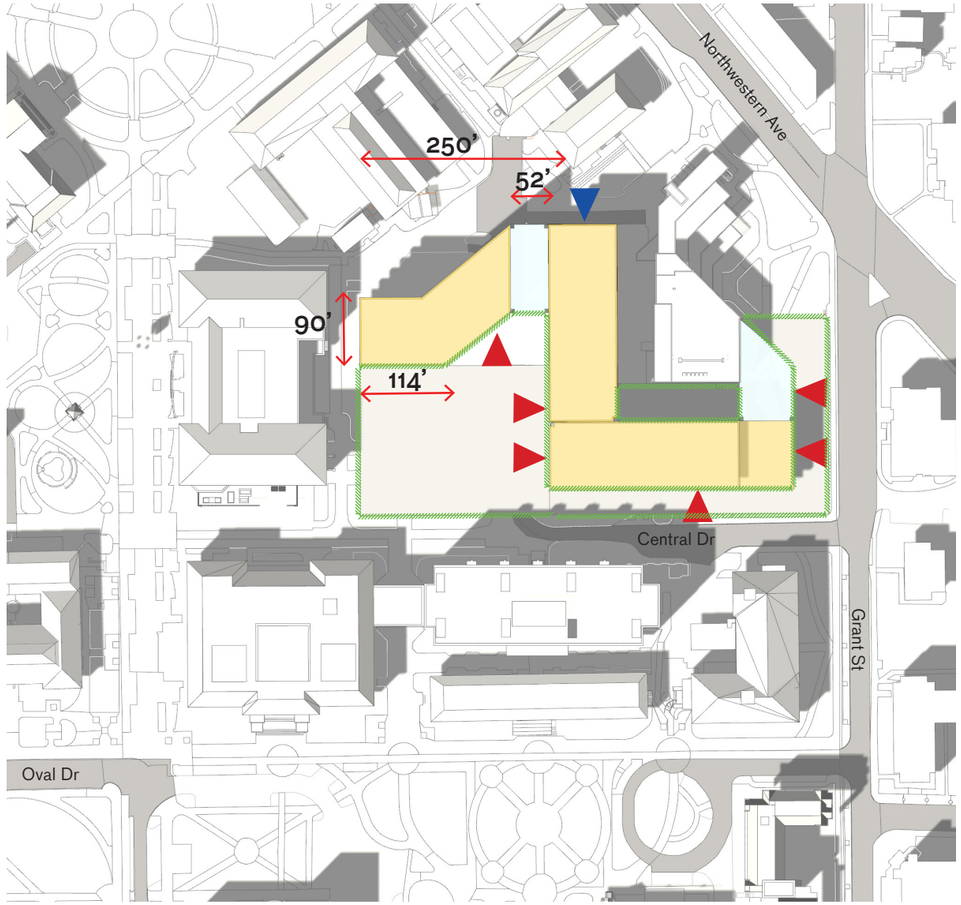
Loading entry

Landscaping zone

Pedestrian entry

Serpentine: Phase 3





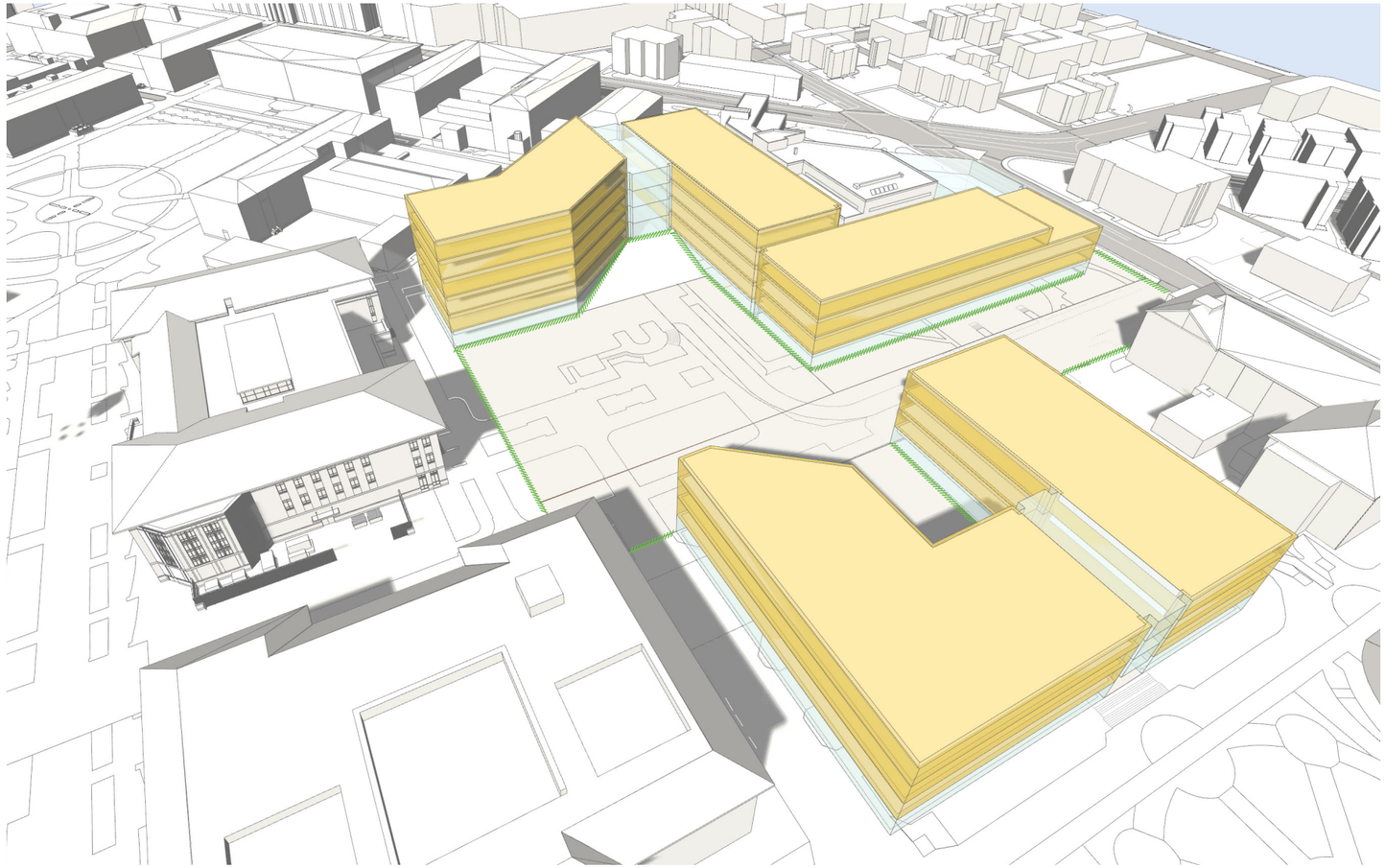
Floors: B+6
GSF: 172,565

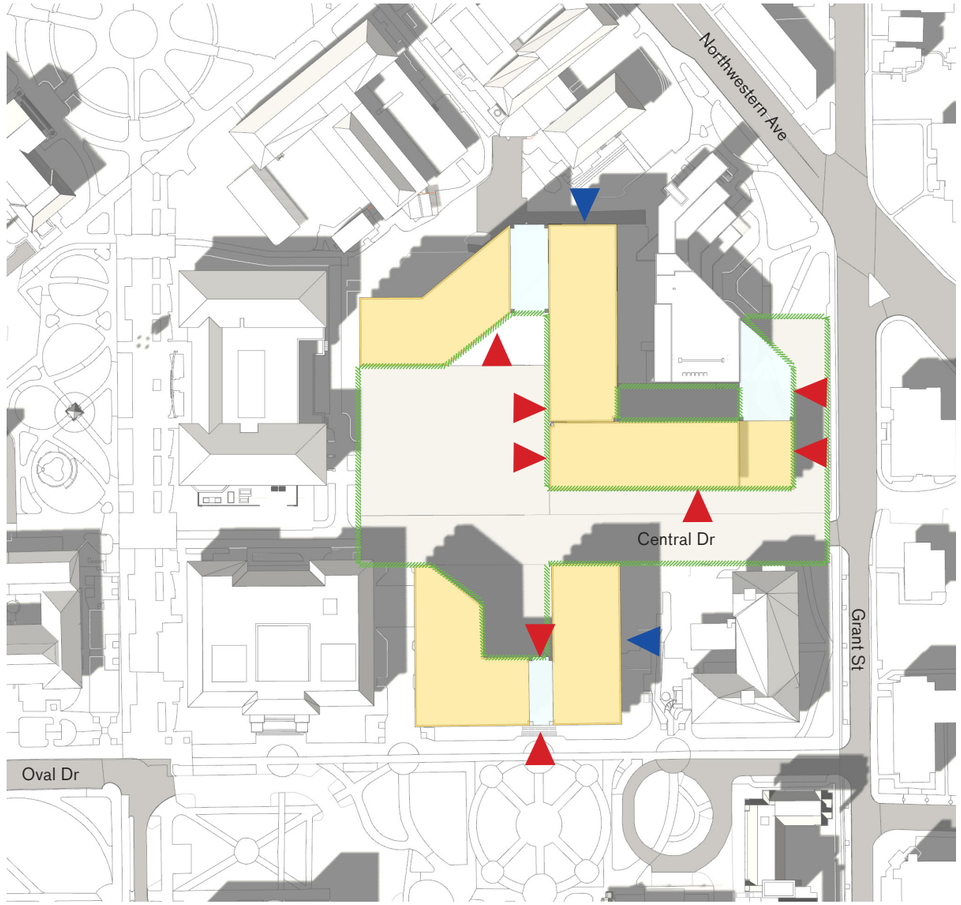
	Phase 1	Phase 2	Phase 3
B	36,591	23,362	20,505
1	36,591	23,362	26,311
2	36,591	23,362	26,311
3	36,591	23,362	26,311
4	22,521	23,362	26,311
5		23,362	26,311
6			20,505
Totals	**168,885**	**140,172**	**172,565**

GRAND TOTAL
481,622

- Loading dock
- Loading entry
- Landscaping zone
- Pedestrian entry

Serpentine: Phase 4





1 Demolish Brown and Heavilon

2 Shared Surface extended across Central Drive

	Phase 1	Phase 2	Phase 3
B	36,591	23,362	20,505
1	36,591	23,362	26,311
2	36,591	23,362	26,311
3	36,591	23,362	26,311
4	22,521	23,362	26,311
5		23,362	26,311
6			20,505
Totals	**168,885**	**140,172**	**172,565**

GRAND TOTAL
481,622

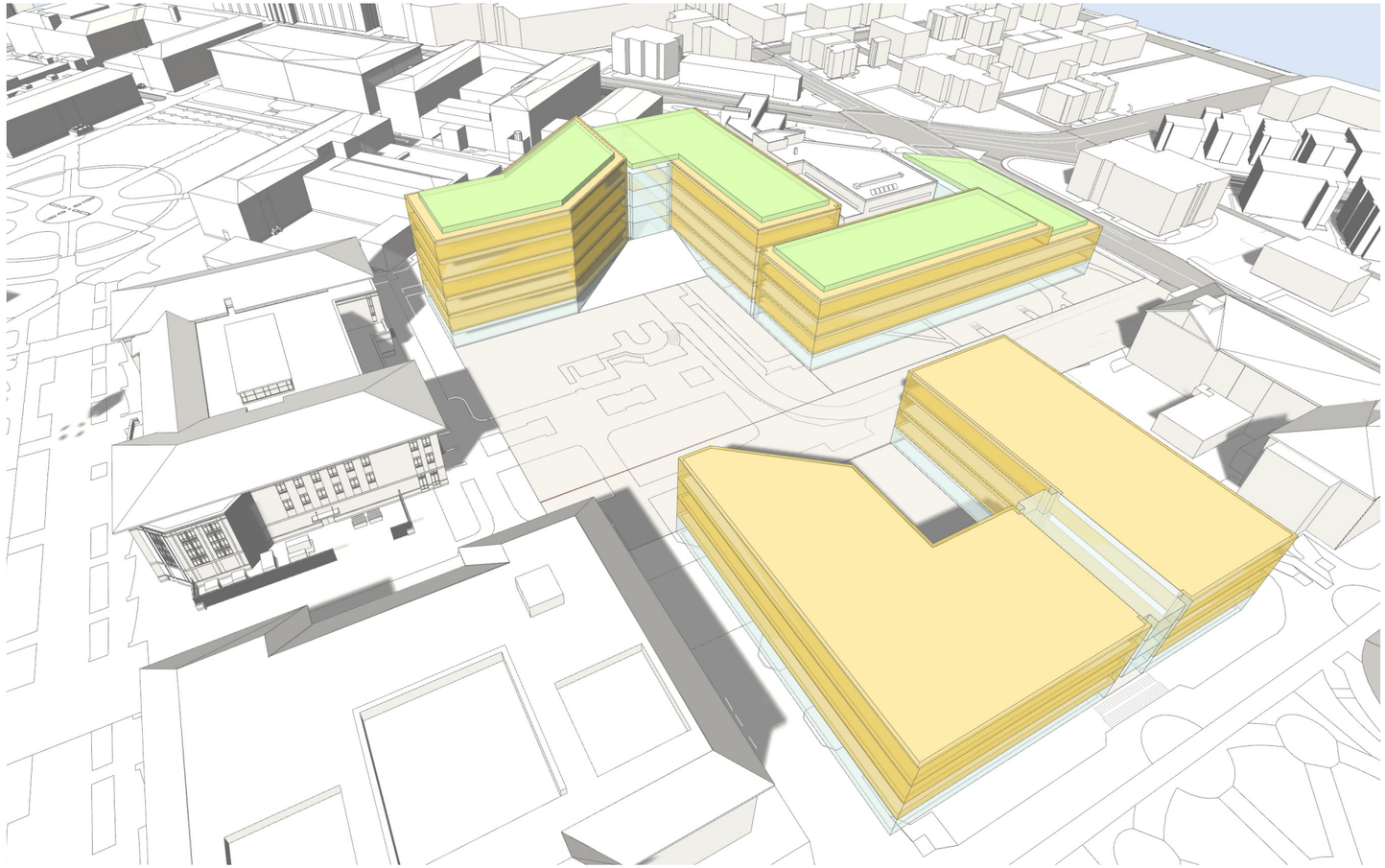
Loading dock

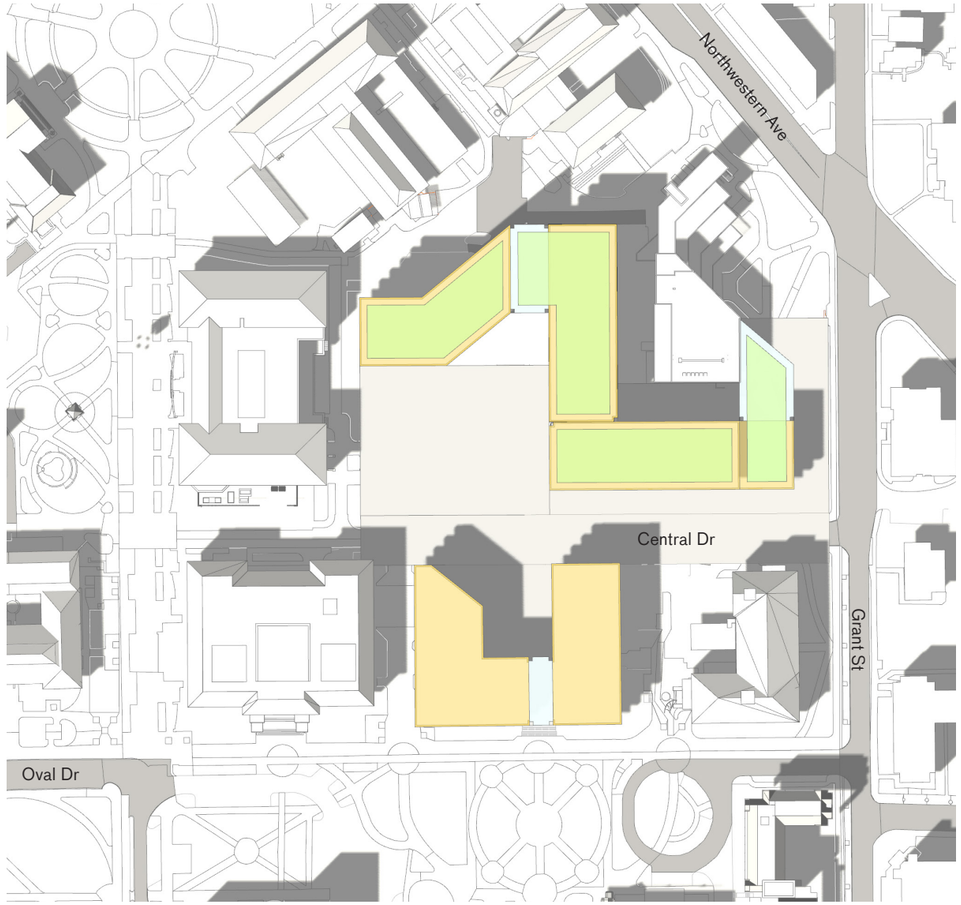
Loading entry

Landscaping zone

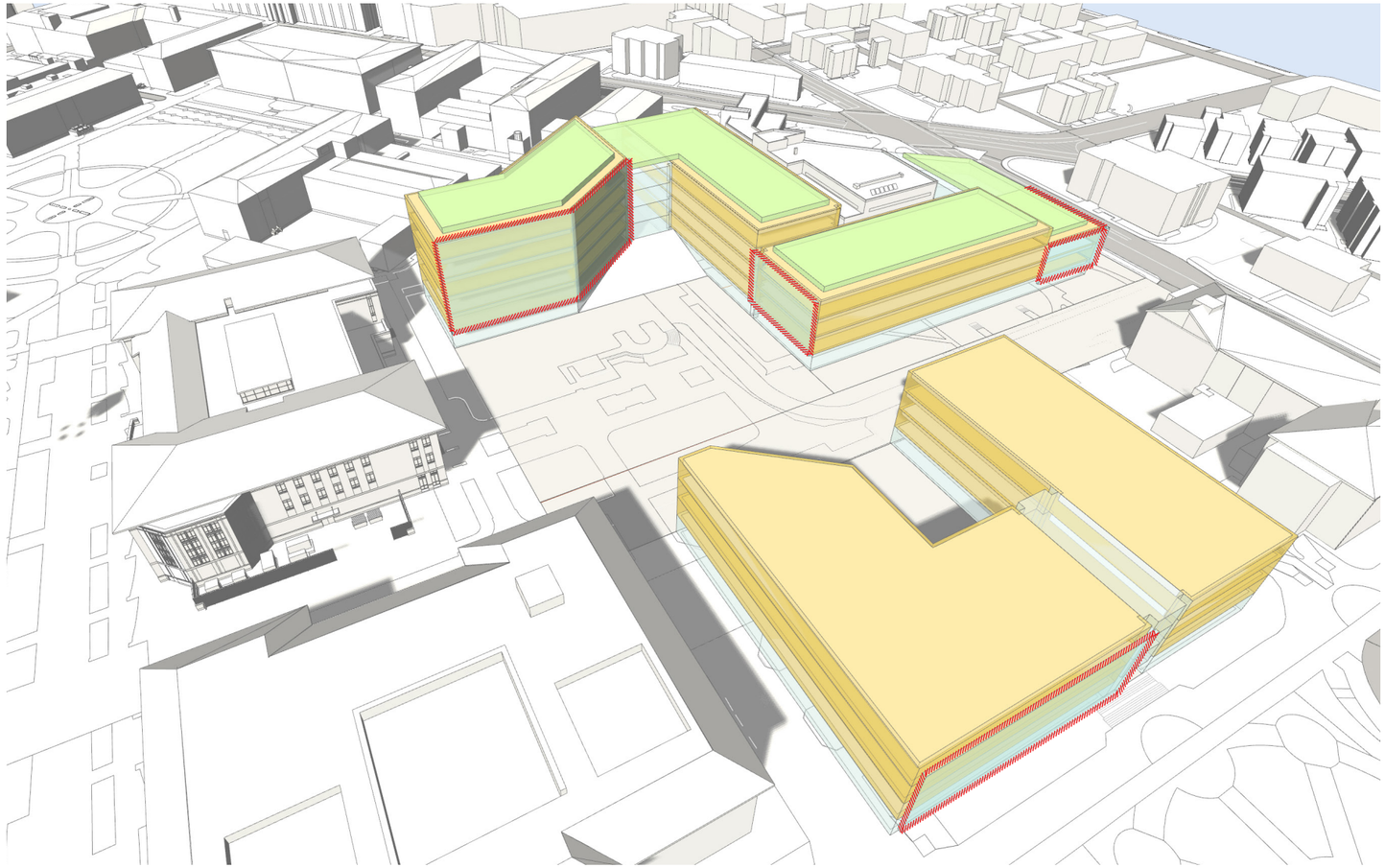
Pedestrian entry

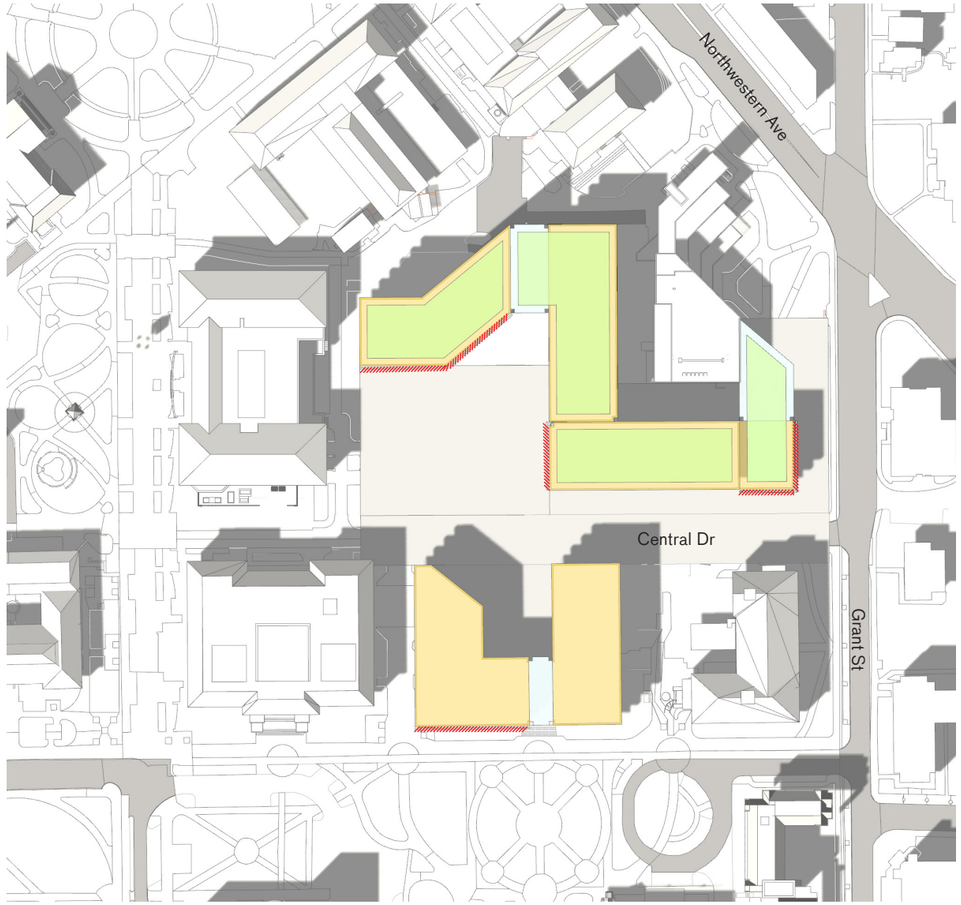
Serpentine: Planted Roofs



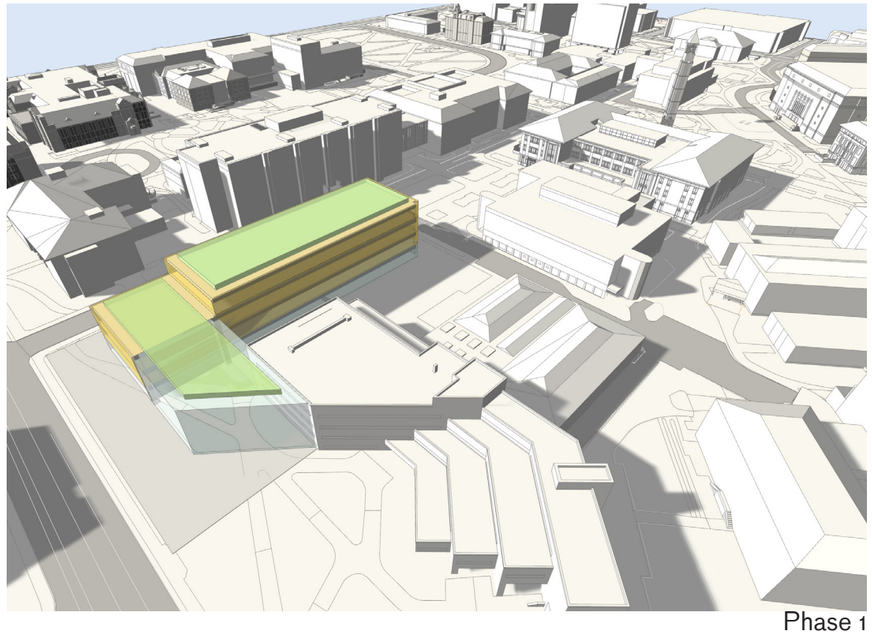


Serpentine: Active and Passive Facades

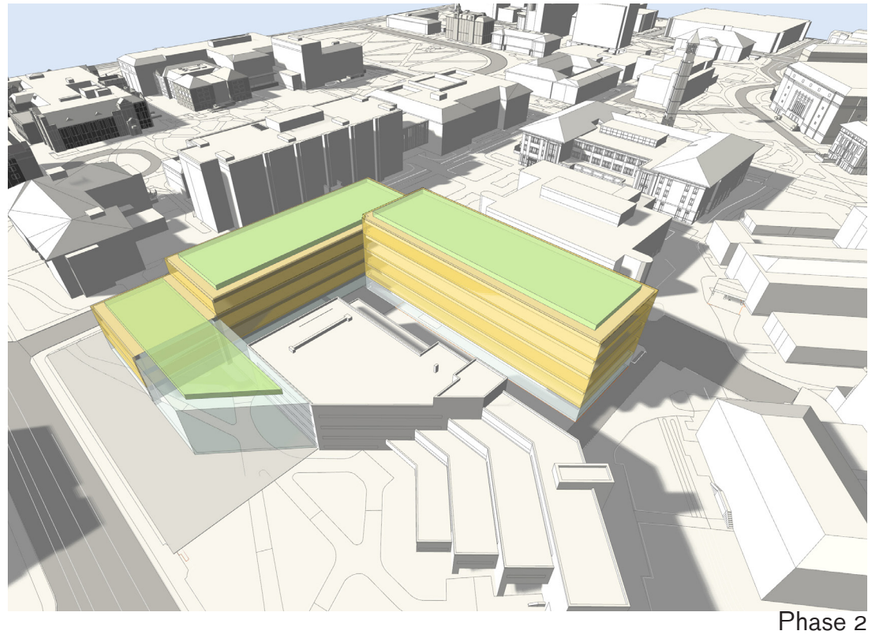




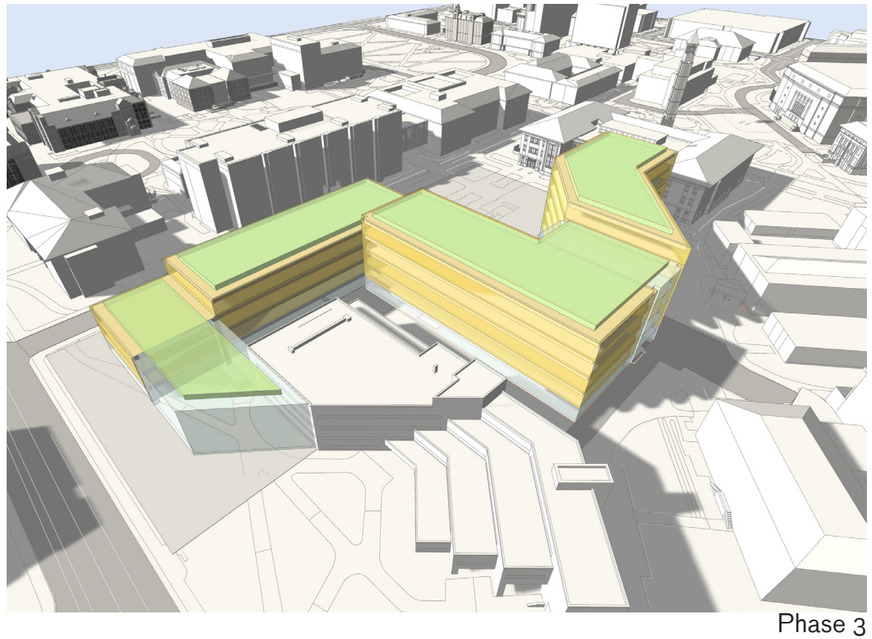
Serpentine: View from North East



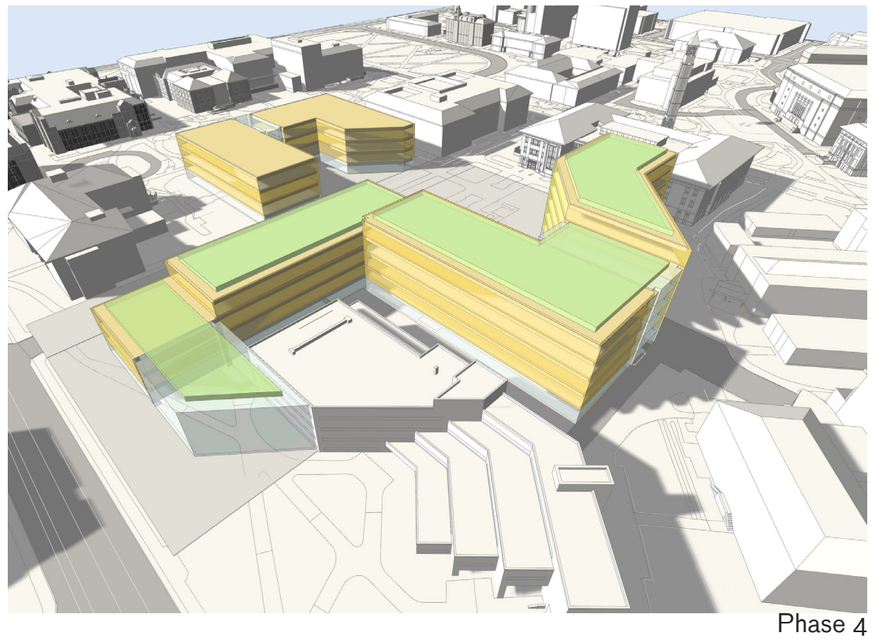
Phase 1



Phase 2

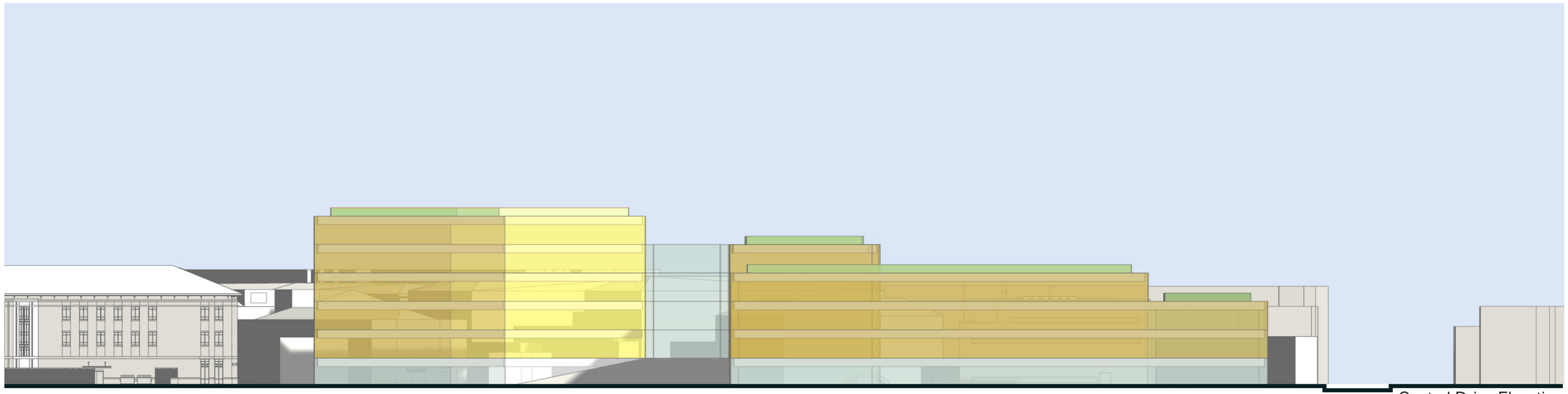


Phase 3

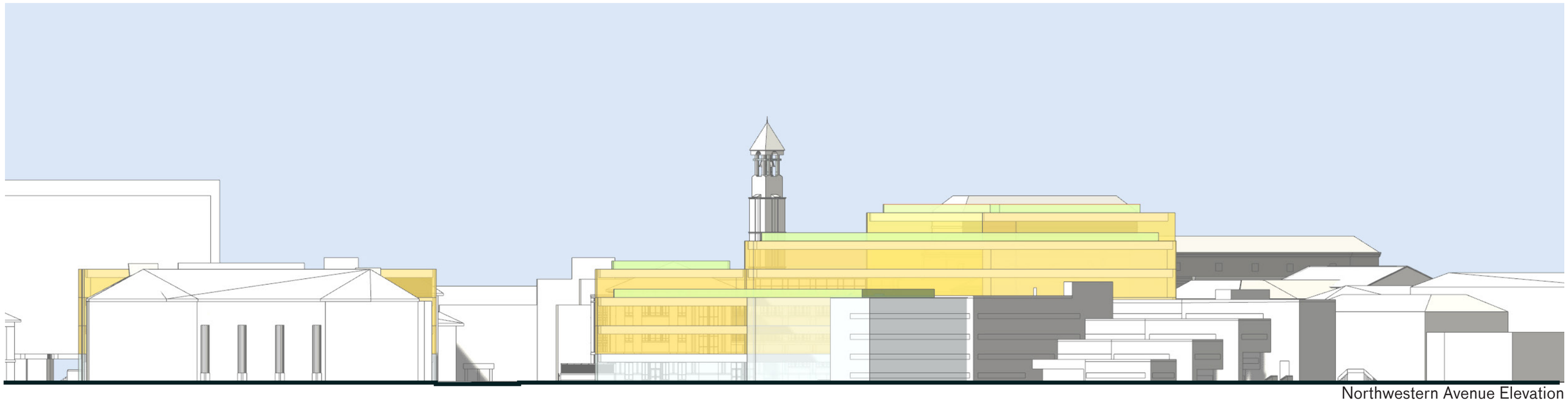


Phase 4

Serpentine: Elevations

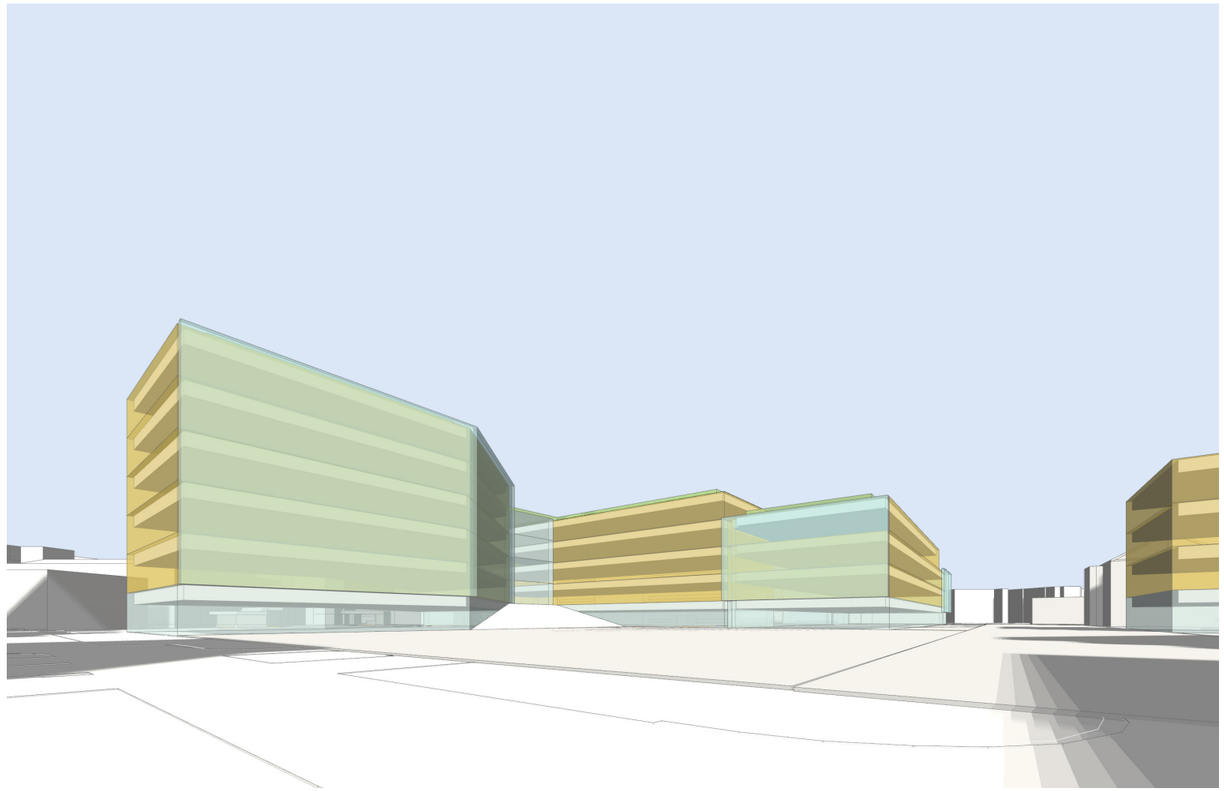


Central Drive Elevation

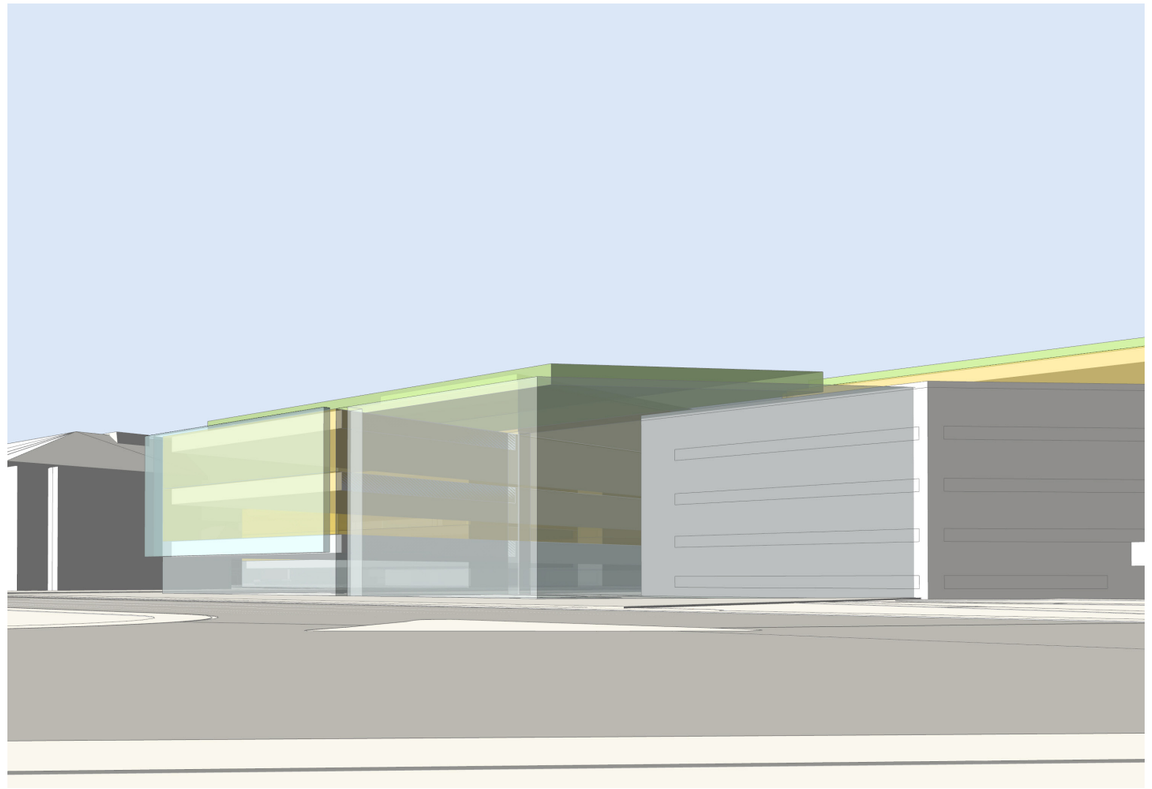


Northwestern Avenue Elevation

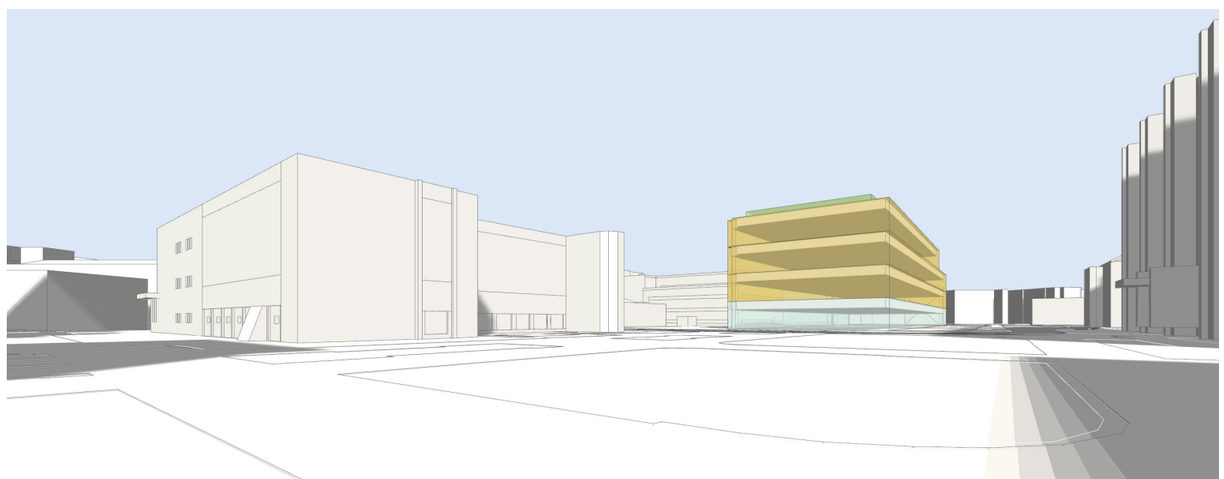
Serpentine: Perspectives



Serpentine: Central Drive, final build out



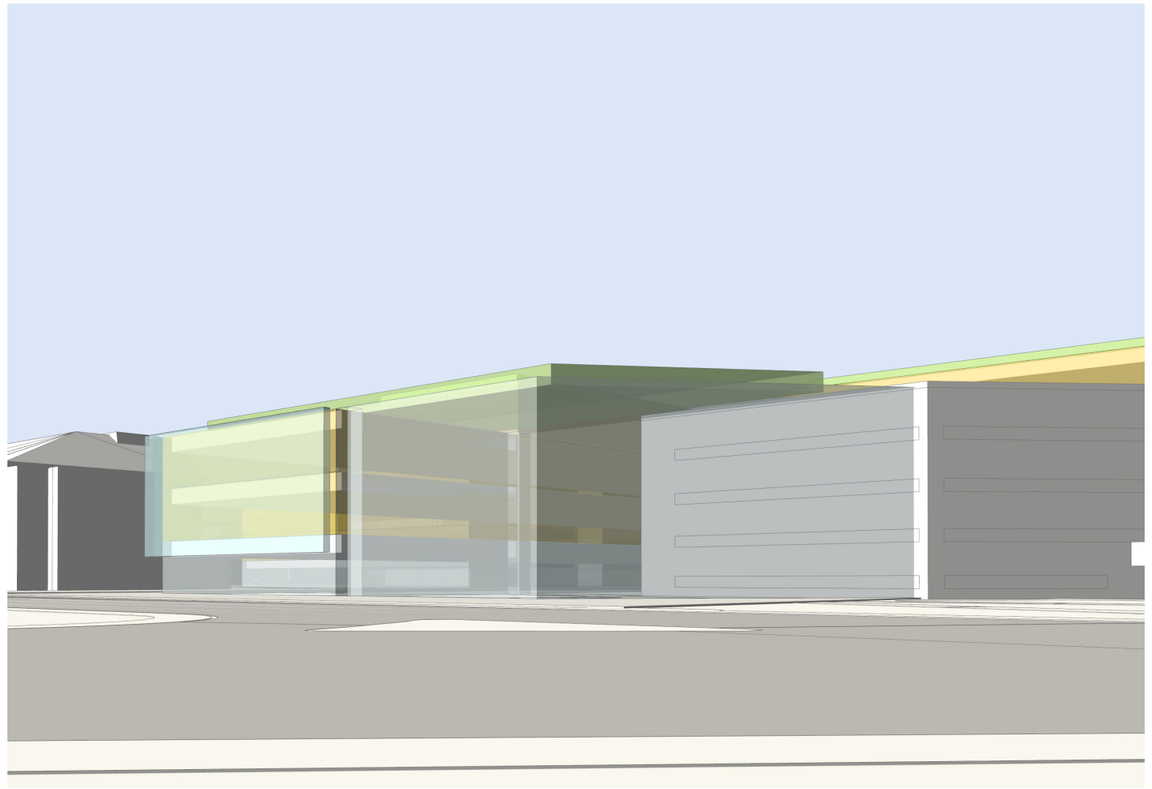
Serpentine: Northwestern Avenue perspective



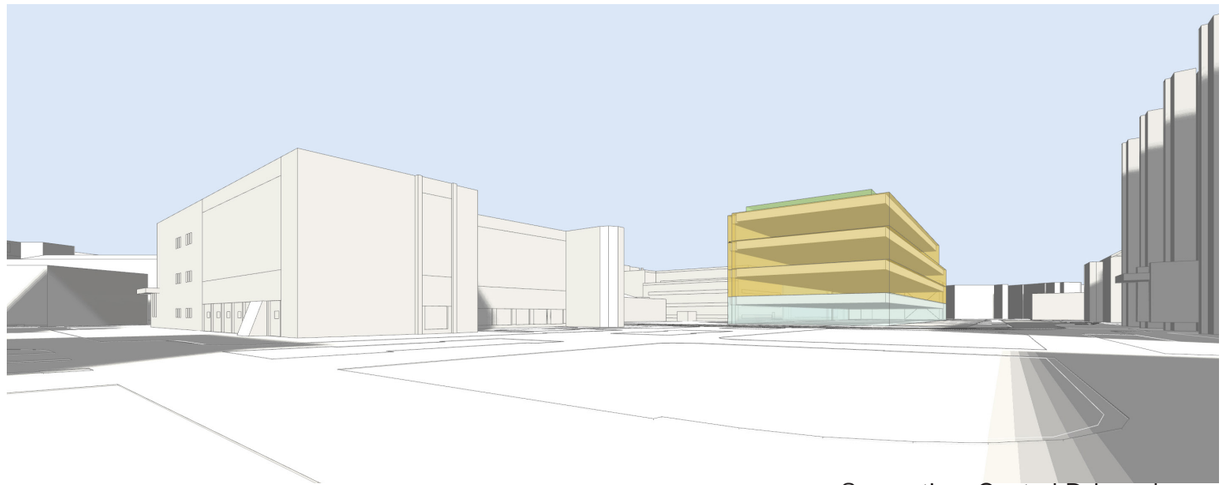
Serpentine: Central Drive, phase 1

Serpentine: Perspectives

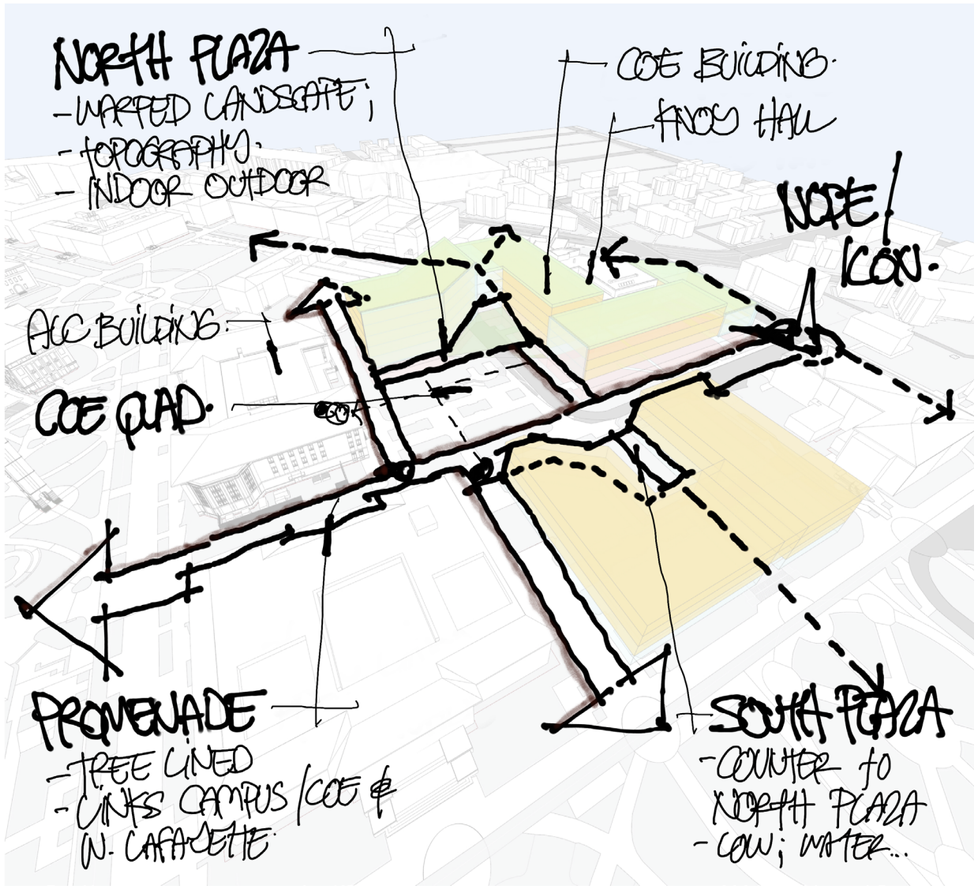
Serpentine: Central Drive, final build out

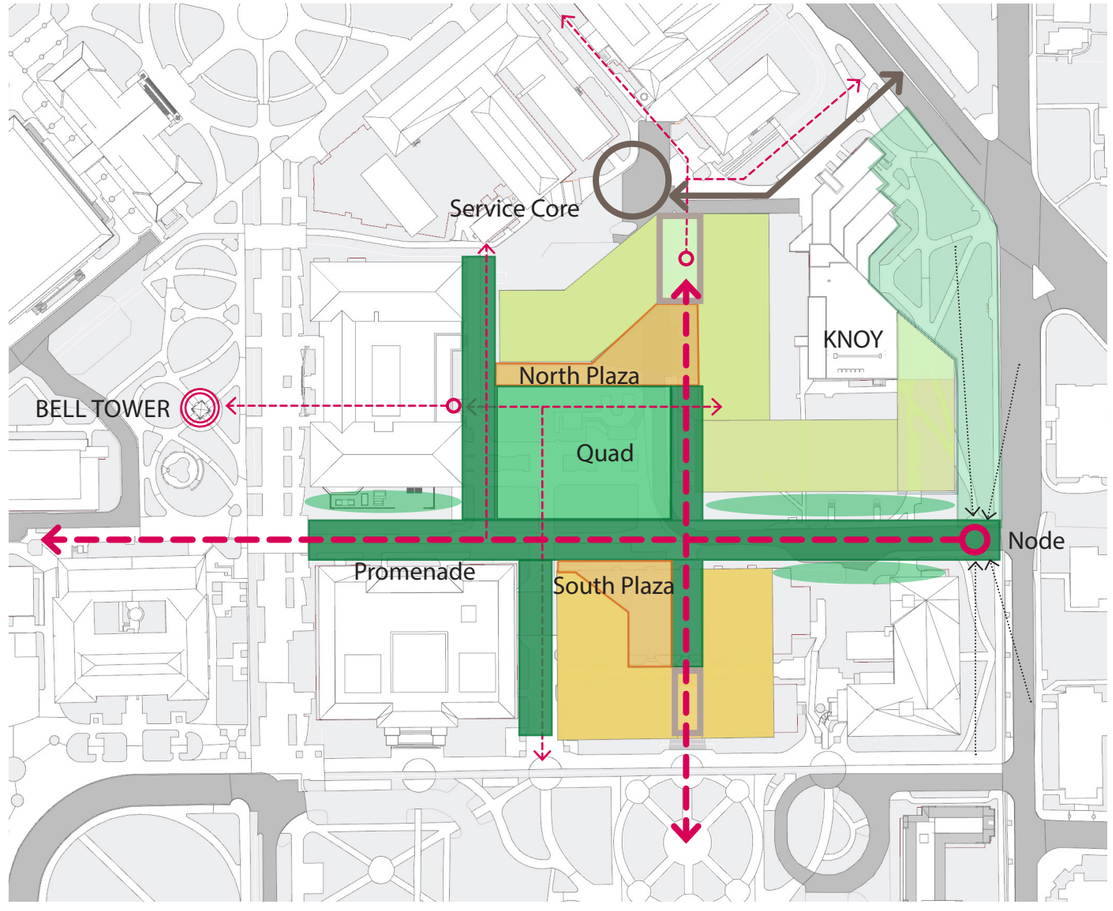


Serpentine: Northwestern Avenue perspective

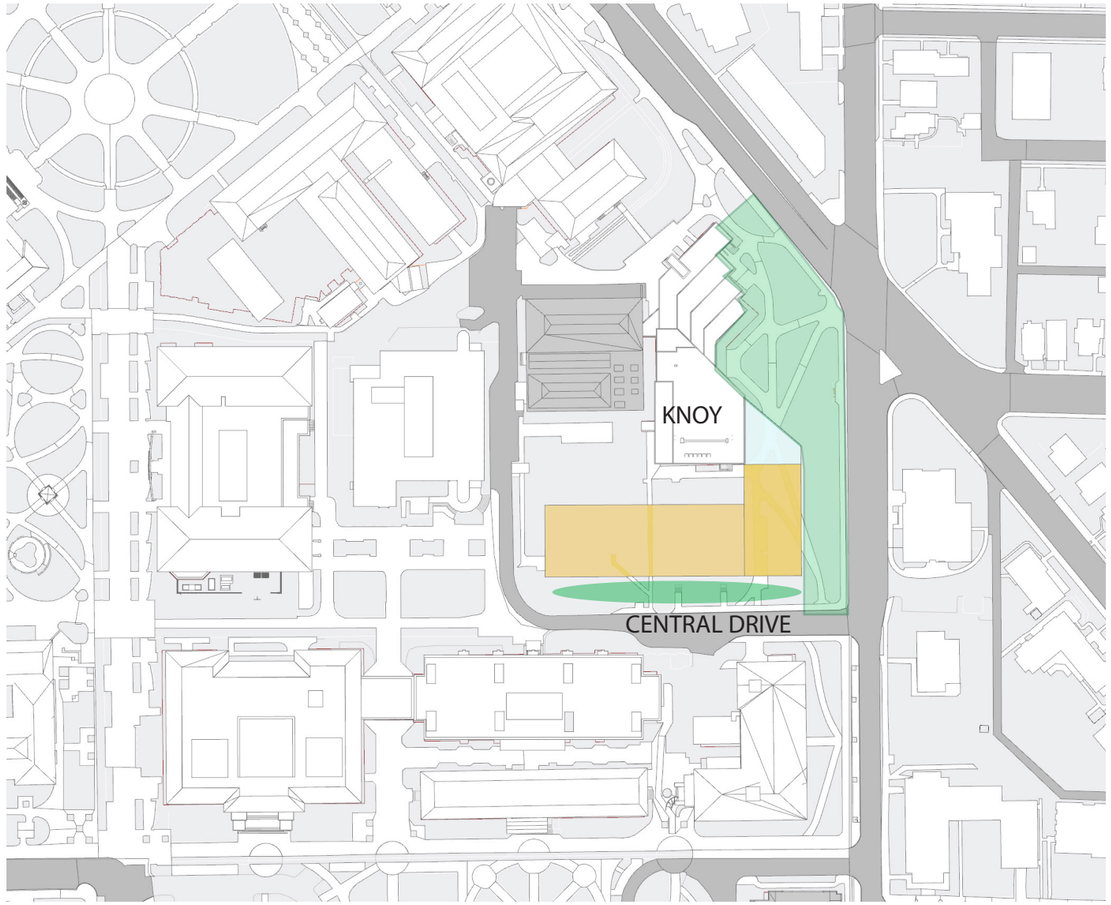


Serpentine: Central Drive, phase 1


NORTH PLAZA
- WARPED LANDSCAPE;
- TOPOGRAPHY.
- INDOOR OUTDOOR
COE BUILDING.
KNOY HALL
NODE/ ICON.
AGC BUILDING.
COE QUAD.
PROMENADE
- TREE LINED
- LINKS CAMPUS / COE &
IN. CAFAYETTE.
SOUTH PLAZA
- COUNTER TO
NORTH PLAZA
- COOL; WATER...


Service Core
KNOY
North Plaza
BELL TOWER
Quad
Node
Promenade
South Plaza

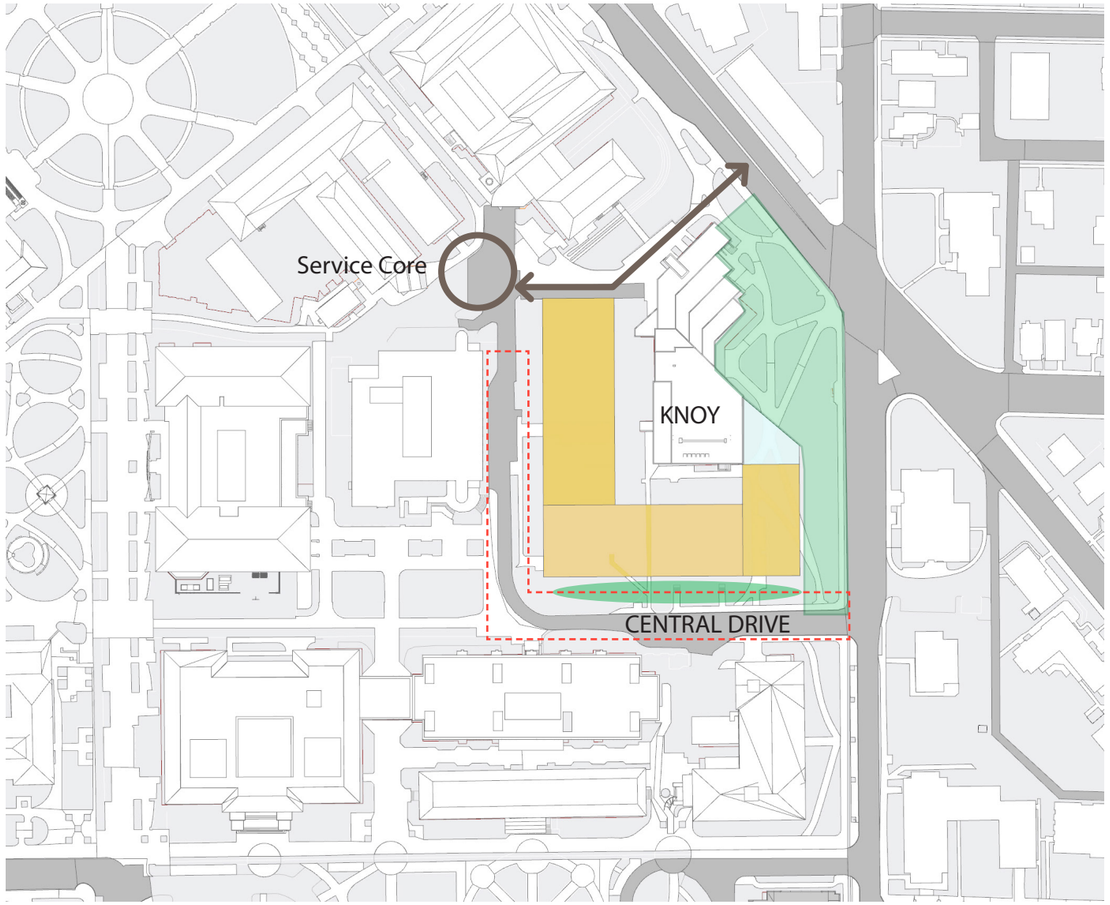
Serpentine: Landscape Phasing Approach



Phase 1

1 Revise Knoy landscape to future condition

2 Establish early Central Drive / Promenade landscape

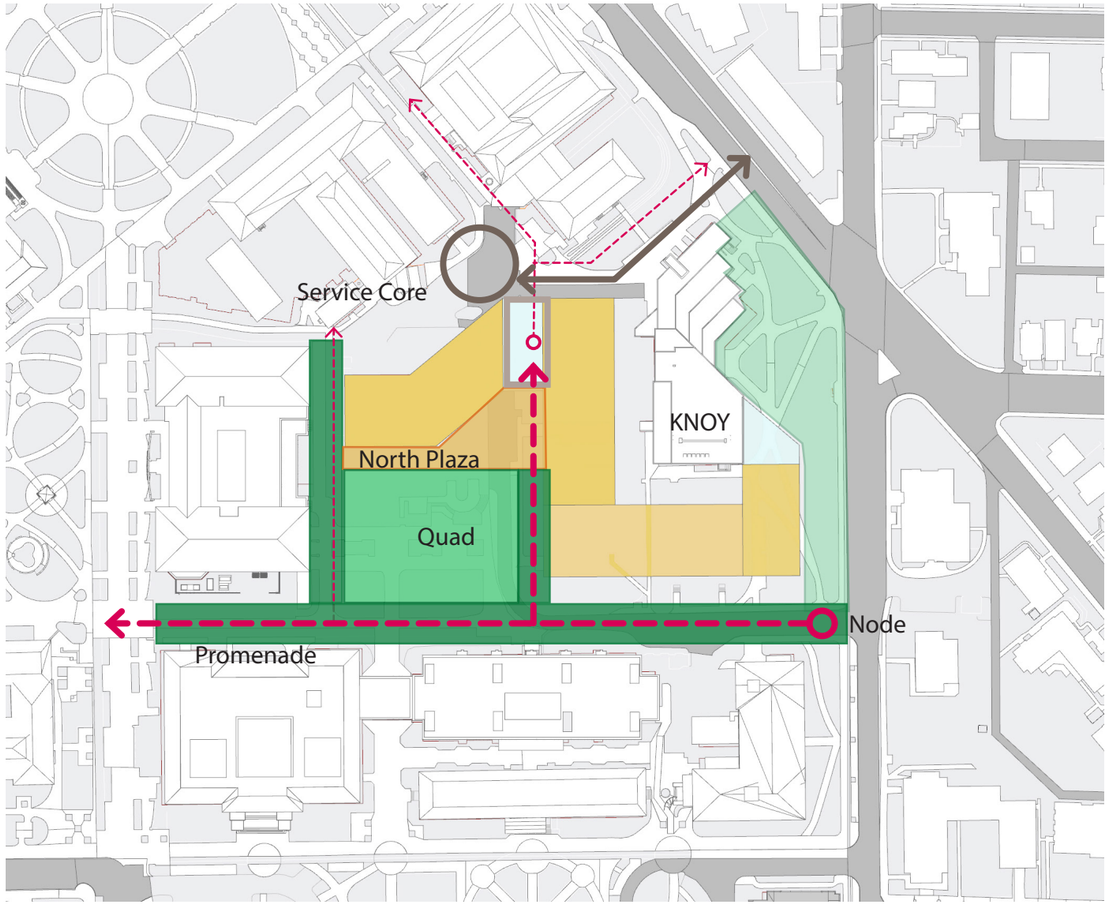


Phase 2

1 Demolish vehicular section of Central Drive

2 Reconnect Service Core to network north of Knoy and CoE

Serpentine: Landscape Phasing Approach



Phase 3

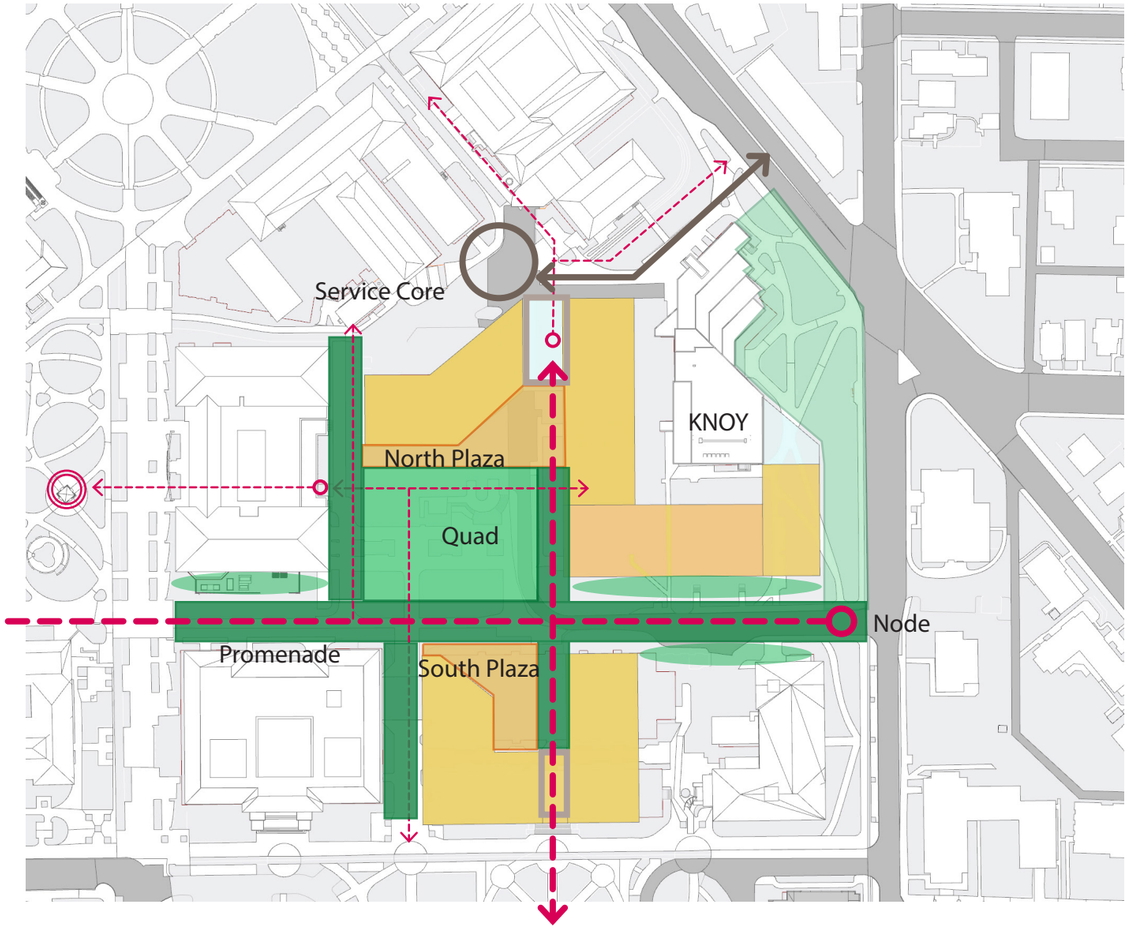
1 Create E-W tree-lined promenade connecting heart of campus to town

2 Announce gateway with landscape node or feature - artwork, curated student work, water, color, light

3 Construct CoE quad and north plaza - warp, create topography

4 Define edge landscape to frame spaces and connect future phases

5 Delineate pedestrian / bike flows to the north



Phase 4

1 Extend edge landscape and pathways to connect new buildings

2 Construct south plaza, a counterpoint to north plaza

3 Enhance landscape in steady state condition

5. Architectural Character

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Architectural Context

Grissom Hall, 1906, 2014

Richard Benbridge Wetherill, 1928-1955

A.A. Potter Engineering Center, 1977

Knoy Hall, 1984

Neil Armstrong Hall, 2007

Wilmeth Active Learning Center, 2017

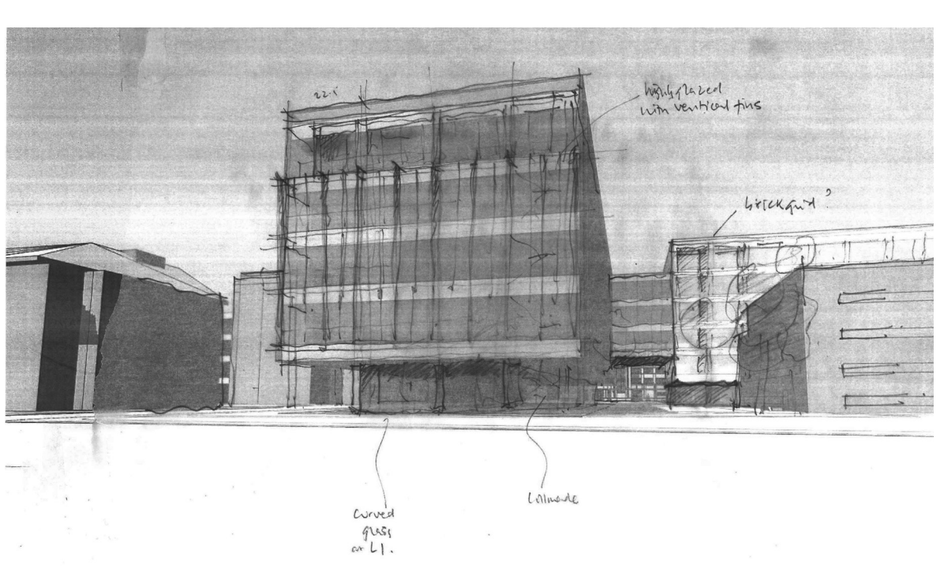
Characteristics

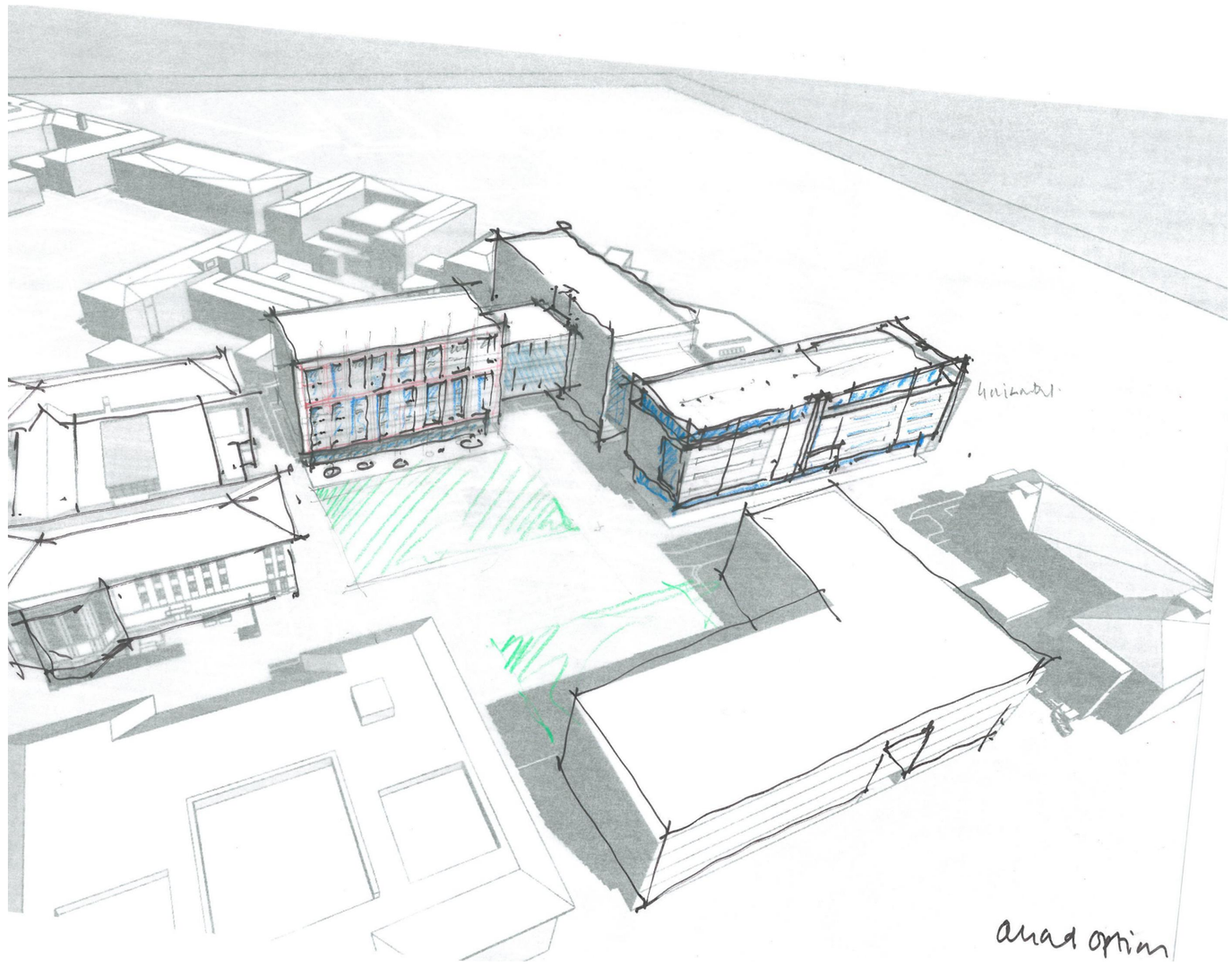
Massing: Horizontal 3-5 story masses broken by vertical elements designating entry or circulation

Materials: Alternating brick panels puncuated by stone or precast concrete

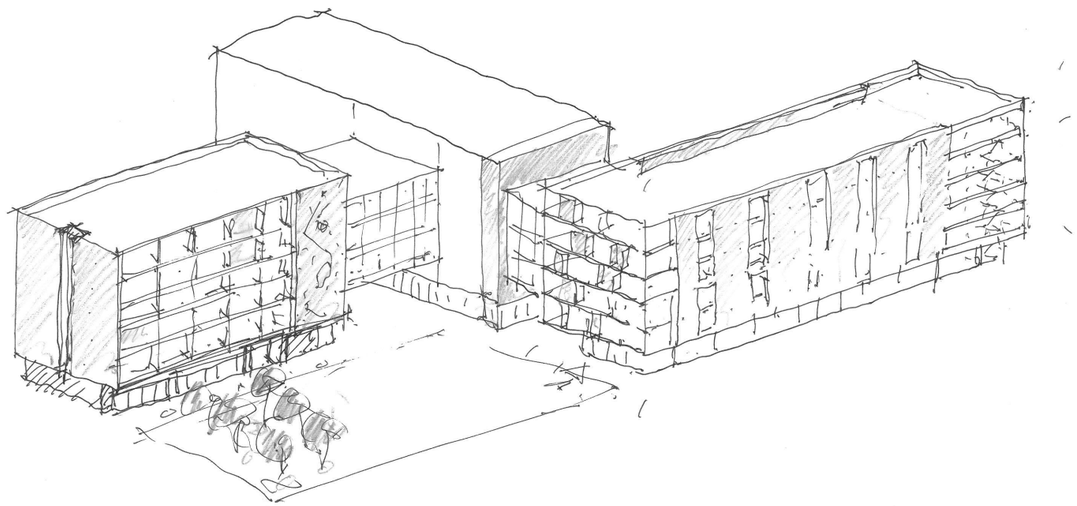
Streetscape: Older buildings provide less street-level transparency

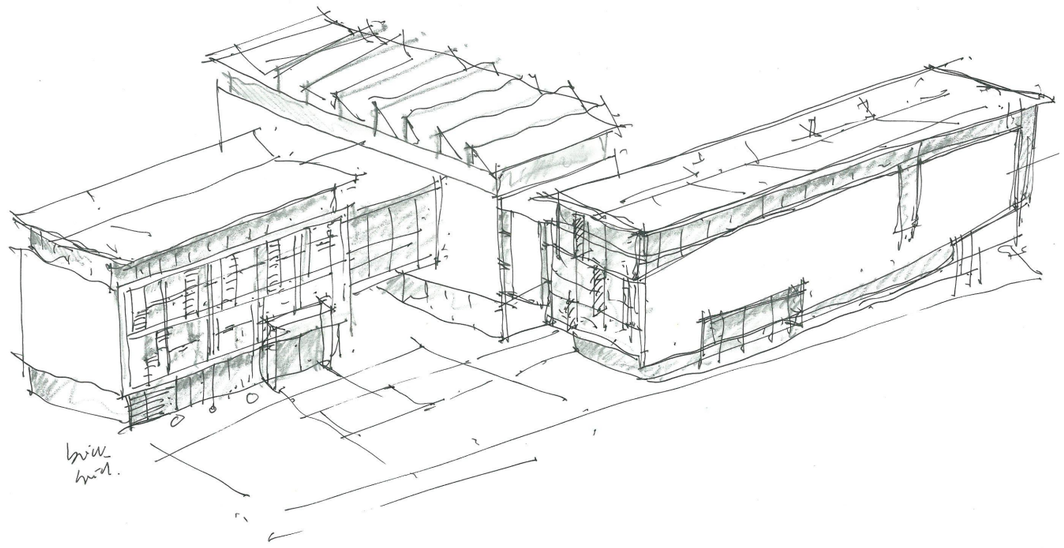
Quad: Sketches



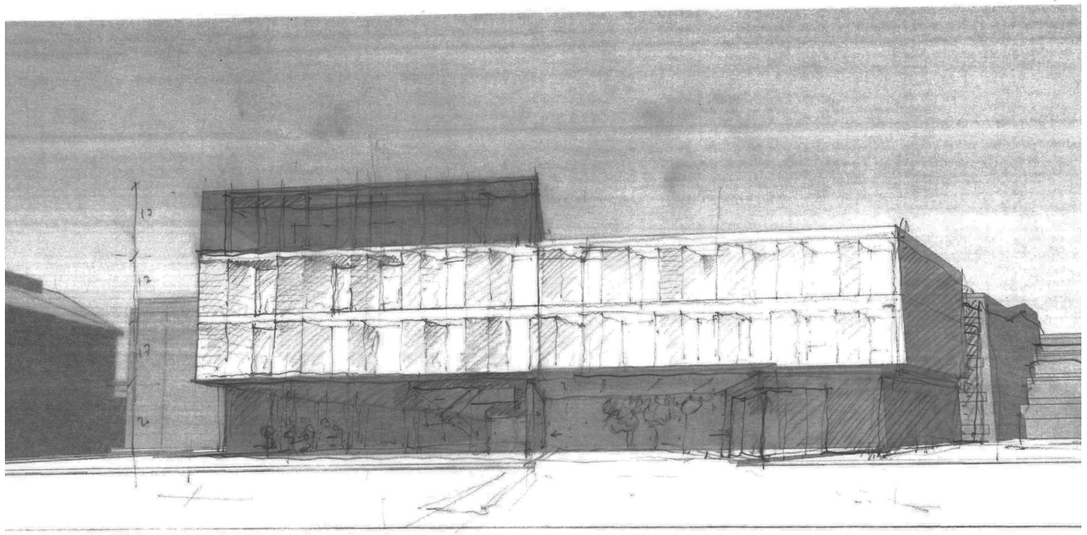


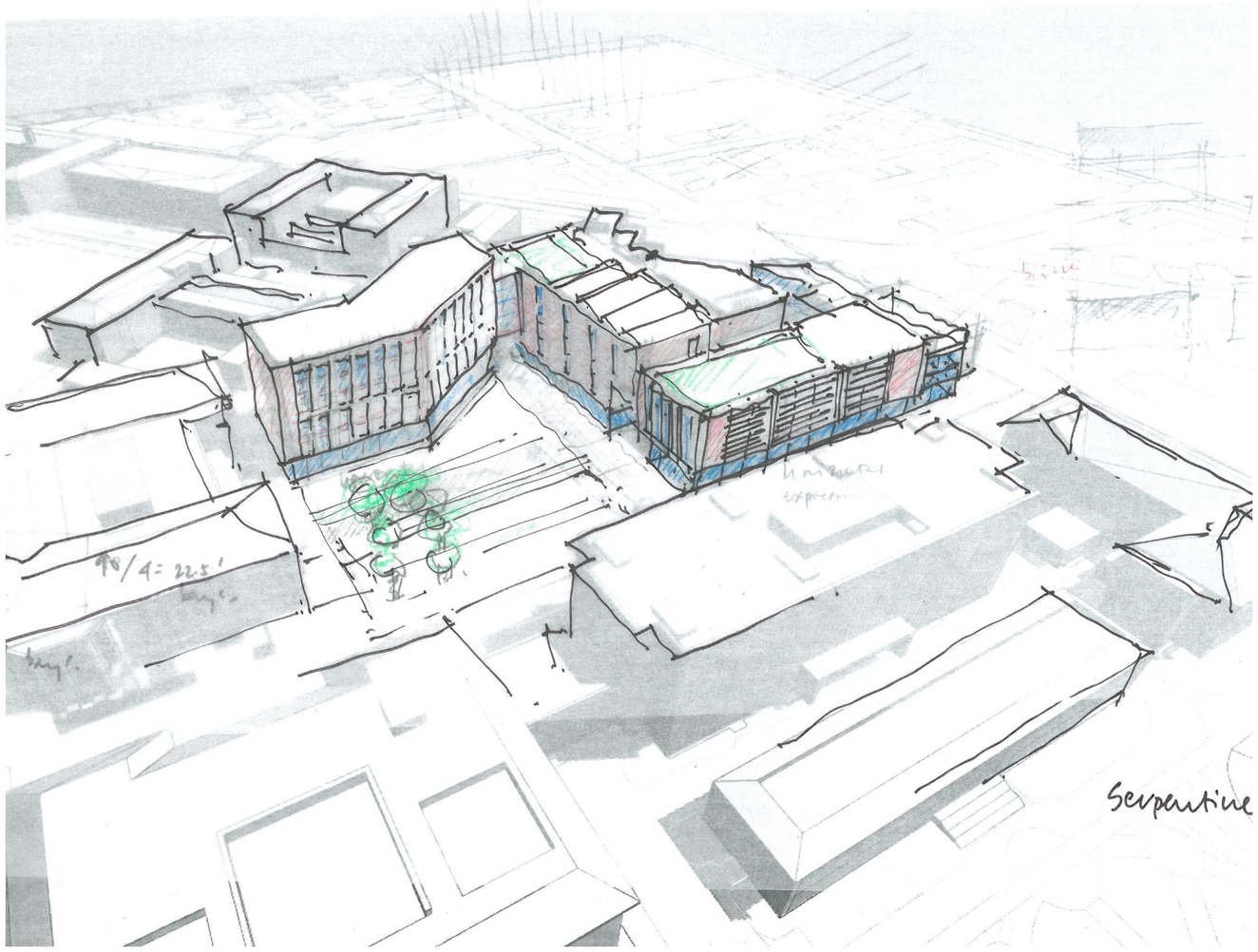
Quad: Sketches

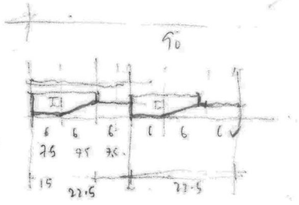




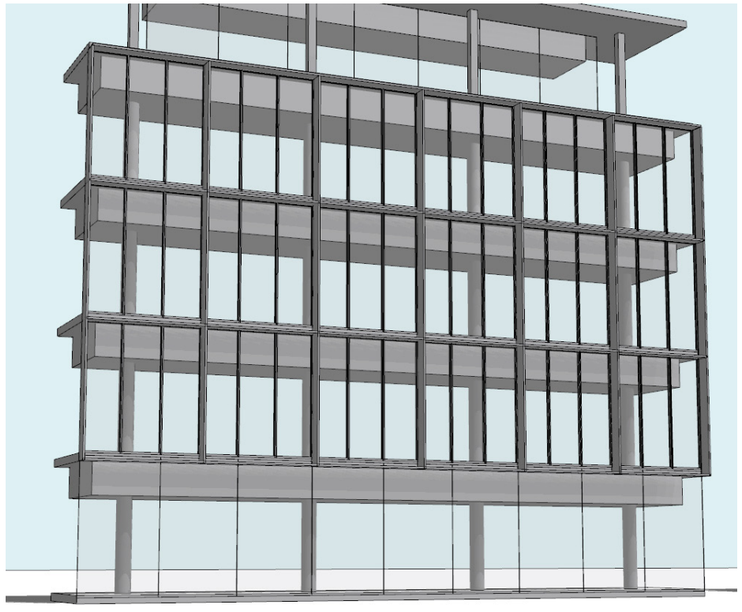
Serpentine: Perspectives

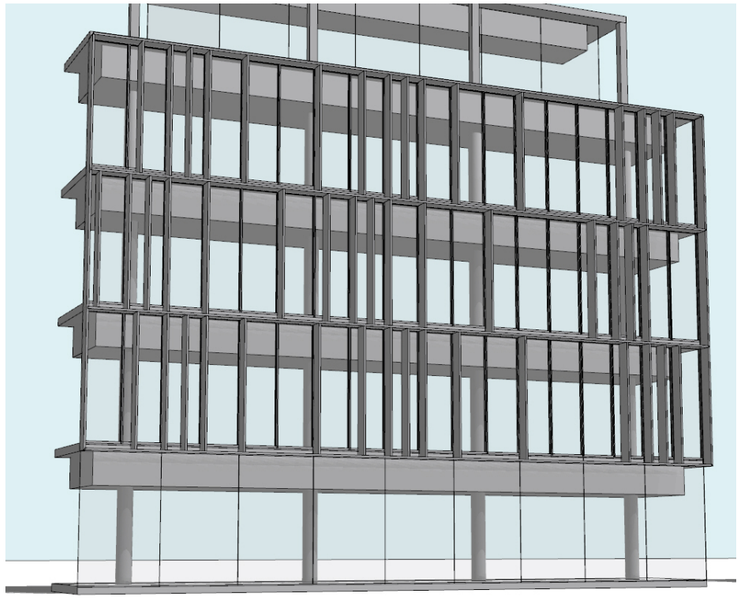






Facade sketches





Brick Precedents

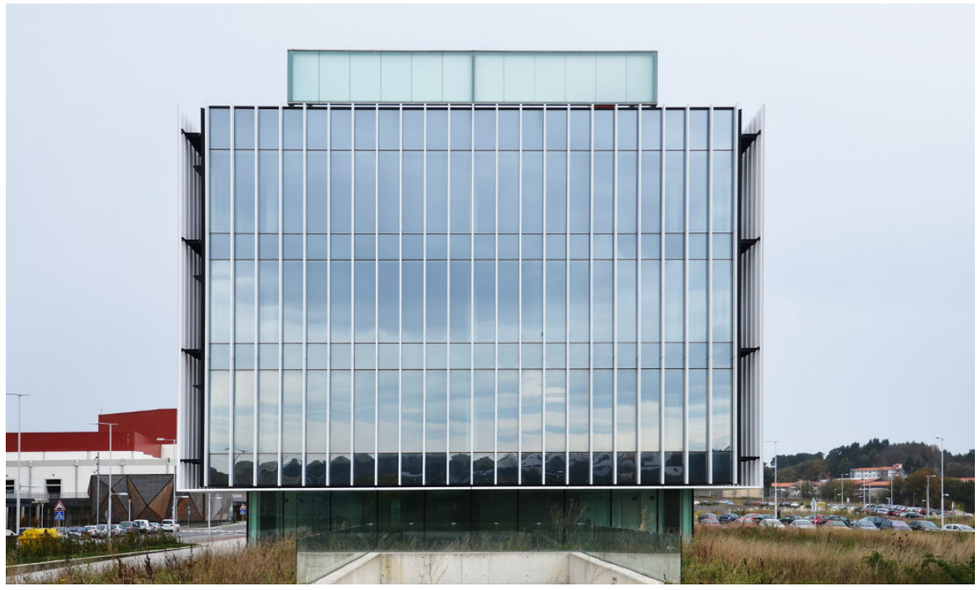
MIT Baker House,Cambridge MA, USA: Alvar Aalto

Polish National Radio Symphony Orchestra, Katowice, Poland: Konior Studio

Dragerwerk, Luebeck, Germany: Max Dudler

Newport Street Gallery, London, UK: Caruso St John

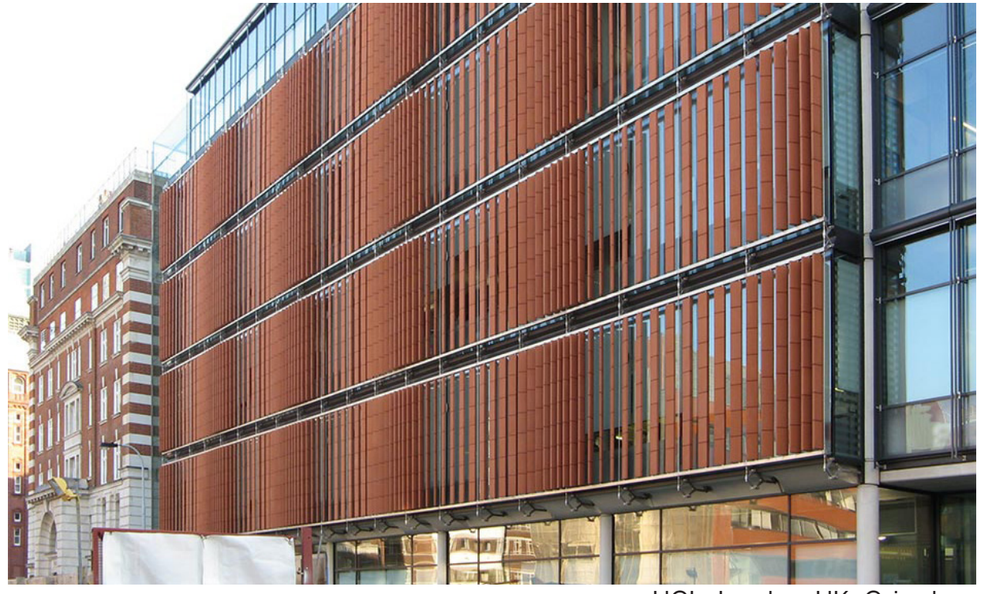
Perforated Metal and Terracotta Shades



Sede del Parque Científicio de la UPV, Blibao, Spain: ACXT

Sede del Parque Científicio de la UPV, Blibao, Spain: ACXT

Seoul Foreign School, Seoul, Korea: Ennead



UCL, London, UK: Grimshaw

Glazed and solid voumes

Goetz Art Collection, Munich, Germany: Herzog & de Meuron

Lisson Art Gallery, London, UK: Tony Fretton

Thank You

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